

WHAT MATTERS FOR REPRESENTATION ALIGNMENT: GLOBAL INFORMATION OR SPATIAL STRUCTURE?

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ABSTRACT

Representation alignment (REPA) guides generative training by distilling representations from a strong, pretrained vision encoder to intermediate diffusion features. We investigate a fundamental question: what aspect of the target representation matters for generation, its *global* semantic information (e.g., measured by ImageNet-1K accuracy) or its spatial structure (i.e. pairwise cosine similarity between patch tokens)? Prevalent wisdom holds that stronger global semantic performance leads to better generation as a target representation. To put this belief to a direct quantitative test, we perform a large-scale empirical analysis spanning 27 different vision encoders and three model scales. The comparison is decisive: across all 27 encoders, spatial structure—not global performance—is what tracks generation quality. Concretely, every spatial-structure metric we consider correlates strongly with generation FID (Pearson $|r| > 0.85$), whereas ImageNet-1K accuracy correlates only weakly ($|r| = 0.26$)—a more than $3\times$ gap in predictive power that holds consistently across model sizes. Building directly on this finding, we introduce two straightforward modifications that specifically accentuate the transfer of *spatial* information: we replace the standard MLP projection layer in REPA with a simple convolution layer and add a spatial normalization layer for the external representation. The payoff is large relative to the change: our method (implemented in <4 lines of code), termed iREPA, consistently improves the convergence speed of REPA—delivering gains on *every* tested vision encoder, model size, and training variant (REPA, REPA-E, Meanflow, JiT, etc.). The consistency and size of these margins motivate revisiting the fundamental working mechanism of representational alignment and how it can be leveraged for improved training of generative models.

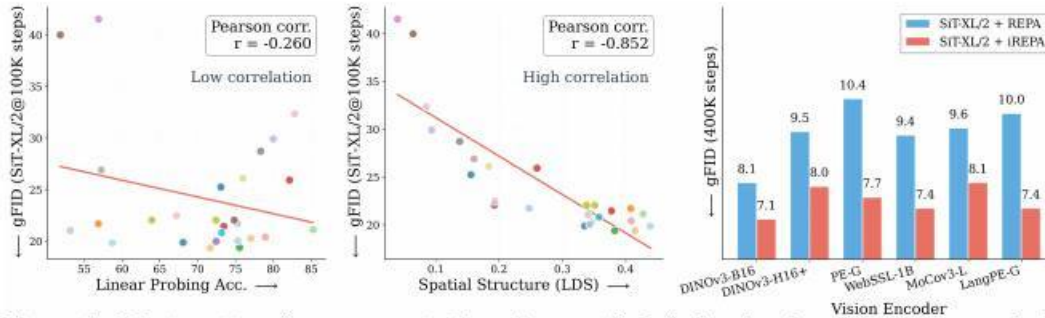


Figure 1: **What matters for representation alignment?** *Left:* head-to-head correlation analysis across 27 diverse vision encoders. Contrary to the prevailing wisdom, spatial structure—not global performance (linear probing accuracy)—is what tracks generation quality, correlating far more strongly with FID than accuracy does. *Right:* acting on this, two simple modifications that accentuate the transfer of spatial features from target representation to diffusion model improve the convergence speed of REPA consistently, and in every setting we test.



Figure 2: **Motivating examples — spatial structure matters.** Two head-to-head pairs in which the accuracy leader is the generation loser—an inverse relationship between ImageNet accuracy and generation quality. *Left:* despite a 29.7-point accuracy lead (82.8% vs. 53.1%), PE-G generates worse than SpatialPE-B (Bolya et al., 2025). *Right:* likewise, WebSSL-1B (Fan et al., 2025) leads on global accuracy (76.0% vs. 53.1%) yet generates worse. **Spatial Self-Similarity:** in both pairs, spatial structure orders the encoders the way generation does—the missing predictor that accuracy gets backwards. See §3 for the spatial structure metric. All results reported at 100K using SiT-XL/2 and REPA.

1 INTRODUCTION

Representation alignment has emerged as a powerful technique for accelerating the training of diffusion transformers (Ma et al., 2024; Peebles & Xie, 2023). By aligning internal diffusion representations with pretrained self-supervised visual encoders, recent methods have demonstrated large, measurable improvements in both convergence speed and final generation quality (Yu et al., 2024; Leng et al., 2025a). Yet despite these empirical successes, the precise mechanism through which self-supervised features enhance diffusion model training remains poorly quantified. This leaves a fundamental question unresolved: is the measured improvement primarily driven by better global semantic information, as commonly quantified through linear probing accuracy, or by better spatial structure, characterized by the pairwise relationships between patch token representations? Answering this quantitatively is the central goal of this paper.

Pinning down this mechanism is crucial for advancing generative model training, as it directly determines which target representation one should select and how much benefit to expect from it. The prevailing assumption is a quantitative one: that an encoder’s usefulness for representation alignment scales with its ImageNet-1K validation accuracy, taken as a proxy for global semantic understanding (Oquab et al., 2024; Chen et al., 2021). That is, target representations with higher ImageNet accuracy are hypothesized to yield proportionally better generation (Yu et al., 2024). Under this view, measured increases in the linear probing accuracy of diffusion features are routinely cited as evidence for the effectiveness of representation alignment, reinforcing global semantic information as the presumed primary driver of the improvement. The following quote from REPA (Yu et al., 2024) captures the current understanding:

“When a diffusion transformer is aligned with a pretrained encoder that offers more semantically meaningful representations (i.e., better linear probing results), the model not only captures better semantics but also exhibits enhanced generation performance, as reflected by improved validation accuracy with linear probing and lower FID scores.”

In this paper, we test this conventional wisdom quantitatively, asking directly which of the two—global semantic information or spatial structure—actually predicts generation quality. Through a controlled, large-scale empirical analysis across 27 diverse vision encoders, including recent large vision foundation models such as WebSSL (Fan et al., 2025), DINOv3 (Siméoni et al., 2025), perceptual encoders (Bolya et al., 2025), and C-RADIO (Heinrich et al., 2024), we arrive at three findings that each run counter to the prevailing assumption.

Higher validation accuracy does not imply a better representation for generation. Contrary to the prevailing assumption, encoders with higher global semantic performance, measured by ImageNet-1K linear probing accuracy, frequently generate *worse*, and the gaps are large. Consider PE-Spatial-B, a spatially-tuned model derived from PE-Core-G (Bolya et al., 2025). Although PE-Spatial-B trails PE-Core-G on patch-token validation accuracy by a wide margin¹ (53.1% vs. 82.8%, a 29.7-point

¹Similar to REPA (Yu et al., 2024), we use linear probing accuracy on patch tokens to measure global semantic performance of external representation as only patch tokens are used for representation alignment.

deficit), it nonetheless produces markedly better generation under REPA (Figure 2). The same reversal appears for WebSSL-1B (Fan et al., 2025), which leads on global accuracy (76.0% vs. 53.1%) yet again lags on generation. As we show in Section §2, this accuracy-generation reversal is not anecdotal: it recurs consistently across a range of target representations, pointing to a systematic principle in how representation alignment benefits diffusion training.

Spatial structure, not global information, determines generation performance. To make this comparison quantitative, we introduce several straightforward metrics for the spatial self-similarity structure (Shechtman & Irani, 2007) between patch tokens (§3). We then run the two candidate predictors—linear probing accuracy and spatial structure—head to head, correlating each against generation FID under REPA across the full set of 27 vision encoders and all 3 model sizes (SiT-B, SiT-L, SiT-XL). The result is a clean separation: every spatial-structure metric correlates strongly with generation FID (Pearson $|r| > 0.852$), while ImageNet-1K validation accuracy correlates only weakly ($|r| = 0.26$)—so spatial structure carries more than $3\times$ the predictive power of global accuracy. This gap also manifests in targeted comparisons: representations carrying almost no global information still yield substantial REPA gains. SAM2, for example, outperforms encoders with $\sim 60\%$ higher ImageNet accuracy (Fig. 3), and even classical spatial descriptors such as SIFT (Lowe, 1999) and HOG (Dalal & Triggs, 2005)—though not the strongest targets—deliver meaningful gains with representation alignment (§3).

Accentuating spatial features improves convergence. Having established spatial structure as the key predictor, we act on it directly with two straightforward modifications (<4 lines of code) that accentuate the transfer of “spatial” information from target representation to diffusion model: (1) a spatial regularization layer that boosts the spatial contrast of the target representations; and (2) a simple convolution layer replacing the standard MLP-based projection (used to map diffusion features to target representation dimensions), which we show discards spatial information (Figure 6a). The empirical return on these minimal changes is consistent and one-directional: our method, termed iREPA, improves the convergence speed of REPA in *every* setting we test—across encoders, across model sizes, and across training recipes *e.g.*, REPA-E (Leng et al., 2025a) and Meanflow (Geng et al., 2025) w/ REPA.

We highlight the main contributions of this paper below:

- Through a large-scale head-to-head analysis over 27 encoders and 3 model scales, we quantify that spatial structure—not global semantic information—is what predicts the effectiveness of representation alignment.
- We introduce a straightforward, fast-to-compute Spatial Structure Metric (SSM) that correlates far more strongly with downstream FID than linear probing accuracy ($|r| > 0.85$ vs. $|r| = 0.26$), giving practitioners a reliable predictor of REPA performance.
- We propose simple modifications that accentuate the transfer of spatial information from the target representation to diffusion features. The resulting method improves convergence speed over REPA *consistently and without exception* across variations in target representation, model architecture, and training recipe (REPA, REPA-E, Meanflow w/ REPA, JiT w/ REPA, etc.).

2 MOTIVATION: GLOBAL INFORMATION MATTERS LESS

We begin with a series of controlled comparisons that isolate global performance (ImageNet-1K accuracy) and show, case by case, that higher global accuracy does *not* translate into better generation with REPA—and often accompanies clearly worse generation. We then show in §3 that these otherwise puzzling observations are cleanly and consistently explained once we instead measure the spatial structure of the target representation.

Recent vision encoders. We compare several recent vision encoders, including Perceptual Encoders (Bolya et al., 2025), WebSSL (Fan et al., 2025), and DINOv3 (Siméoni et al., 2025), pairing a high-accuracy encoder against a lower-accuracy one in each case. Take PE-Spatial-B (80M), a small spatially tuned model derived from PE-Core-G (1.88B) (Bolya et al., 2025). Despite a 29.7-point advantage in ImageNet-1K accuracy for the larger model (82.8% vs. 53.1%), the accuracy leader is the generation loser: as target representations for REPA, PE-Core-G reaches only FID 32.3 while PE-Spatial-B reaches FID 21.0—a 35% relative improvement in favor of the *lower*-accuracy encoder

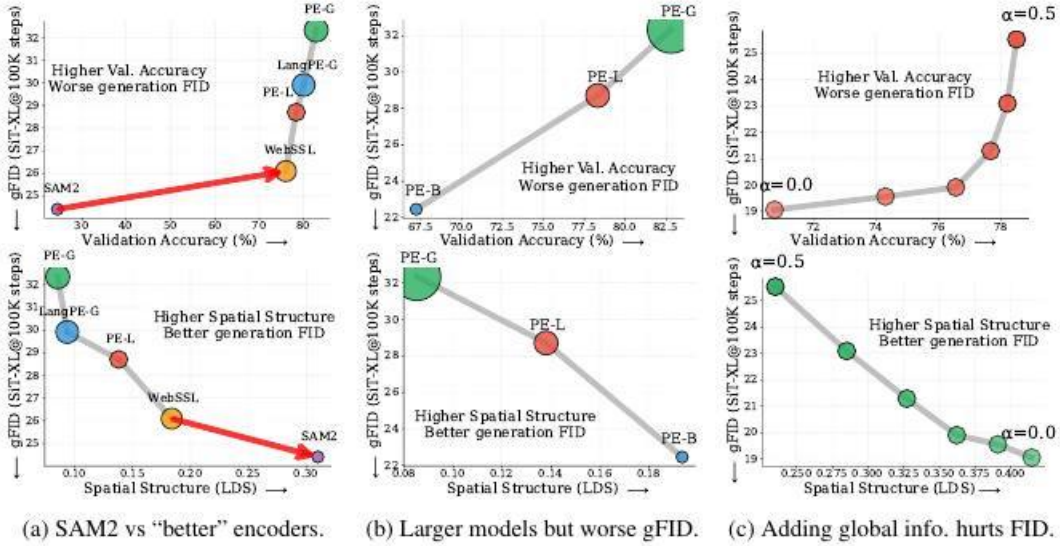


Figure 3: **Higher global information does not imply better REPA performance.** **Top row:** three independent controlled comparisons in which global performance fails to track generation FID under REPA. (a) SAM2-S, at only 24.7% validation accuracy, generates better than encoders with $\sim 60\%$ higher accuracy. (b) Within an encoder family, the larger, higher-accuracy variants generate the same or worse. (c) Sweeping in CLS-token global information raises accuracy while FID gets strictly worse. **Bottom row:** the *same* three cases replotted against spatial structure, which orders them the way generation does—so spatial structure, not global performance, is the reliable indicator. See §3 for the large-scale analysis across spatial-structure metrics. All results are reported w/o classifier free guidance, SiT-XL/2 w/ REPA and 250 NFE (Yu et al., 2024) for inference.

(Figure 2). WebSSL-1B (1.2B) tells the same story: it leads on ImageNet-1K accuracy (76.0% vs. 53.1%) but again generates worse (FID 26.1 vs. 21.0).

SAM2 outperforms vision encoders with much higher ImageNet-1K accuracy. To push the comparison to an extreme, we use the SAM2-S vision encoder (46M) (Ravi et al., 2024)—a small model with almost no global information, at only 24.1% validation accuracy (Fig. 3a). Even from this very low starting point, SAM2-S delivers better REPA FID than encoders carrying far more global information, *e.g.*, PE-Core-G at 82.8% accuracy—an encoder that leads SAM2-S by nearly 60 accuracy points yet trails it on generation.

Larger models within the same encoder family can generate similarly or worse. A common expectation is that scaling up within an encoder family buys better representations, as measured by ImageNet-1K accuracy. For representation alignment, however, this expected monotonic gain does not materialize: larger variants generate about the same (DINOv2) or, in several families, distinctly *worse* (PE, Cradio) despite their higher accuracy (Fig. 3b). (Yu et al., 2024) report the same non-improvement for DINOv2 and attribute it to “we hypothesize is due to all DINOv2 models being distilled from the DINOv2-g model and thus sharing similar representations”. We show in §3 that spatial structure accounts for this trend directly.

Adding more global information can hurt generation. We next control the amount of global information directly, injecting global semantics via the CLS token into local patch tokens (with DINOv2 as encoder) and sweeping the mixing strength. The CLS token mixing operation is $\mathbf{p}_i^{\text{new}} = \mathbf{p}_i + \alpha \cdot \mathbf{c}$, where $\mathbf{p}_i$ denotes the i^{th} patch token, $\mathbf{c}$ the CLS token, and $\alpha \in [0, 0.5]$ controls the mixing strength. The two metrics move in opposite directions as α increases from 0 to 0.5 (Figure 3c): linear probing accuracy rises monotonically from 70.7% to 78.5% (a +7.8-point gain), while generation quality degrades in lockstep, with FID worsening from 19.2 at $\alpha = 0$ to 25.4 at $\alpha = 0.5$ (a +6.2-point, $\sim 32\%$ relative regression). More global information here buys more accuracy and strictly worse generation.

Taken together, these four comparisons point in one direction: the global performance of a representation is not a reliable indicator of its REPA performance. We next show (§3) that spatial structure provides the missing predictive signal.

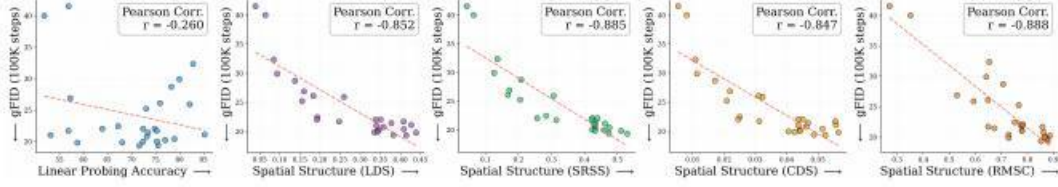


Figure 4: **Spatial structure correlates far more strongly with generation quality than linear probing.** Head-to-head correlation with FID across 27 diverse vision encoders (SiT-XL/2, REPA). Linear probing is barely predictive (Pearson $|r| = 0.260$), whereas all four spatial-structure metrics are strongly predictive—LDS ($|r| = 0.852$), CDS ($|r| = 0.847$), SRSS ($|r| = 0.885$), and RMSC ($|r| = 0.888$)—a roughly $3.3\times$ advantage that is consistent across every metric rather than driven by any single one. See Fig. 10 for detailed plots with encoder labels.

- **Finding 1.** Better global semantic information (e.g., ImageNet-1K accuracy) does not imply a better representation for generation.

3 SPATIAL STRUCTURE MATTERS MORE

We hypothesize that spatial structure, rather than global information, is what drives the generation performance of a target representation. To test this quantitatively, we define several straightforward metrics of the spatial self-similarity structure (Shechtman & Irani, 2007) of target representations. We then show two things with numbers: (i) every spatial-structure metric correlates far more strongly with gFID than ImageNet-1K accuracy does, and (ii) these same metrics resolve each of the previously unexplained observations from §2.

Measuring spatial self-similarity structure. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we define:

- *Self-similarity.* Let $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ be the extract patch representations, with $T = H \times W$ patches. Let kernel $K_{\mathcal{X}}(\cdot, \cdot) \in \mathbb{R}$ measure appearance self-similarity (Shechtman & Irani, 2007) between patch tokens. Here, we use the cosine kernel $K_{\mathcal{X}}(t, t') = \frac{(\mathbf{x}_t, \mathbf{x}_{t'})}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2}$.
- *Spatial distance.* Let $\mathcal{P} \in \mathbb{N}^{T \times 2}$ be the spatial location of each of the T tokens in coordinate space and $d(\cdot, \cdot) \in \mathbb{N}$ be the Manhattan distance between pairs of tokens.

We then measure *spatial* self-similarity metric as how self-similarity $K_{\mathcal{X}}$ varies with lattice distance d between patch tokens. Intuitively, larger values indicate stronger spatial organization (closer patches more similar to each other than patches further away). By default, we use a simple correlogram contrast (local vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}, \mathcal{P}) = \mathbb{E}_{d(t, t') \in (0, r_{\text{near}})} K_{\mathcal{X}}(t, t') - \mathbb{E}_{d(t, t') \geq r_{\text{far}}} K_{\mathcal{X}}(t, t').$$

The final spatial self-similarity score (LDS) is computed as the expectation over patch representation $\mathcal{X}$. We use $r_{\text{near}} = r_{\text{far}} = H/2$ here, though we found correlation to be robust to their exact choices. Please see Appendix B for exact details and alternative spatial metrics (CDS, SRSS, RMSC) which also perform effectively (see Fig. 4).

Spatial structure correlates far more strongly with generation performance than linear probing. We run the two predictors head to head across 27 diverse vision encoders. The verdict is unambiguous (Figure 4): the commonly used linear probing accuracy is barely predictive of generation FID (Pearson $|r| = 0.26$), whereas every SSM metric is strongly predictive (Pearson $|r| > 0.85$)—more than $3\times$ the correlation strength, and consistent across all four spatial metrics rather than an artifact of any single one.

Generalization across model scales. This ranking is not specific to one model size. Repeating the analysis for SiT-B, SiT-L, and SiT-XL (Figure 5), linear probing stays weakly correlated at every scale ($|r| < 0.306$), while spatial structure stays strongly correlated throughout ($|r| > 0.826$)—so the predictive advantage of spatial structure persists uniformly across the model-size axis.

- **Finding 2.** Spatial structure correlates much higher with generation performance than linear probing. We next use spatial metrics to explain previously unexplained trends.

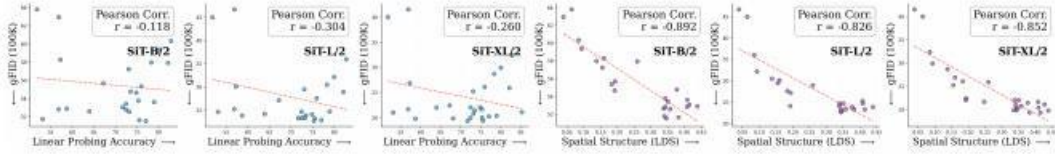


Figure 5: **Correlation analysis across model scales.** At every model scale, spatial structure (right) correlates with gFID far more strongly than linear probing (left)—the ranking of the two predictors never flips with model size, so the advantage of spatial structure is scale-invariant.

Spatial structure explains previously unexplained trends. As shown in §2, ImageNet-1K accuracy fails as a predictor of representation-alignment effectiveness. We now check whether the spatial metrics succeed on exactly the four cases where accuracy failed—and in each case the direction of the spatial metric matches the direction of generation quality.

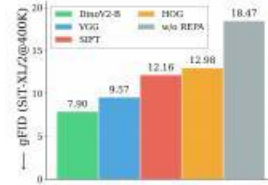
(1) *PE-Spatial-B vs PE-Core-G*: as Figure 2 shows, PE-Core-G leads by 29.7 accuracy points (82.8% vs. 53.1%) yet generates worse (FID 32.3 vs. 22.0). Spatial structure reconciles this: PE-Spatial-B has substantially higher spatial structure than PE-Core-G (Figure 2), correctly predicting the observed generation ranking that accuracy gets backwards.

(2) *SAM2 outperforms “better” vision encoders*: despite its very low 24.1% accuracy, SAM2 beats encoders with $\sim 60\%$ higher accuracy on generation (§2). Its stronger spatial structure directly accounts for this reversal (Fig. 3a).

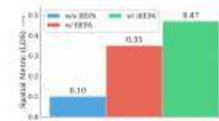
(3) *Larger models in the same encoder family underperform*: larger variants gain accuracy but lose spatial structure, and it is the spatial-structure drop—not the accuracy gain—that matches their equal-or-worse REPA generation (Fig. 3b). This is consistent with the similar-or-worse scaling behavior reported by Yu et al. (2024).

(4) *Adding global information to patch tokens via the CLS token hurts generation*: mixing in the CLS token raises global accuracy but lowers spatial contrast among patch tokens (Figure 3c), causing tokens to become spuriously similar to unrelated ones (e.g., foreground object to background). The reduced spatial structure tracks the observed generation regression—again matching FID where accuracy points the wrong way.

If spatial structure matters more, can we use SIFT or HOG features for REPA? Yes—and this is a strong test of the claim, since these features carry essentially no learned global semantics. While not the strongest targets, classical spatial features such as SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), and intermediate VGG features (Simonyan & Zisserman, 2014) each still yield measurable REPA gains, providing direct evidence that spatial structure alone is sufficient to benefit representation alignment, with no additional global information required.



Can we use spatial metrics to explain gains with REPA? Yes. As shown, the introduced spatial metrics account for both the gains from standard REPA and the additional gains from our improved training recipe (iREPA), which we introduce and quantify next in §4.



4 iREPA: IMPROVING REPRESENTATION ALIGNMENT BY ACCENTUATING WHAT MATTERS

Having established quantitatively that spatial structure is the key predictor, we now turn the finding into a training recipe and measure the payoff. We introduce two straightforward modifications to the original REPA recipe, both aimed at enhancing the transfer of spatial features from the teacher (vision encoder) to the student (diffusion transformer), and evaluate the resulting improvements against REPA under matched conditions throughout.

Convolutional projection layer instead of MLP. The standard REPA recipe maps diffusion feature dimensions to those of the external representation with a 3-layer MLP projection. As Figure 6a shows, this projection is lossy: it visibly diminishes the spatial contrast between patch tokens. We therefore

replace the MLP with a single lightweight convolutional layer (kernel size 3, padding 1) that operates directly on the spatial grid, whose inductive bias preserves local spatial relationships by construction.

Spatial normalization layer. As in Siméoni et al. (2025), we observe that the patch tokens of pretrained vision encoders carry a sizable global component, quantified by the high linear probing accuracy of the *mean* of patch tokens (Figure 14). Figure 6b makes the trade-off concrete: mixing this global information (mean of patch tokens) back into the patch tokens raises global performance but lowers the spatial contrast between individual tokens, leaving foreground tokens (e.g., an object) spuriously similar to unrelated ones (e.g., background).

The quantitative ranking from §3 makes the choice clear: since spatial structure, not global information, predicts generation, we deliberately trade away this global component (mean of patch tokens) to sharpen the spatial contrast between patch tokens. The sharper contrast yields a cleaner spatial signal (pairwise similarity between patch tokens) and, in turn, better REPA performance. Concretely, we add a simple spatial normalization layer (Ulyanov et al., 2016) to the patch tokens of the target representation:

$$\mathbf{y} = \frac{\mathbf{x} - \gamma \mathbb{E}[\mathbf{x}]}{\sqrt{\text{Var}[\mathbf{x}] + \epsilon}},$$

where $\mathbf{x} \in \mathbb{R}^{B \times T \times D}$ represents the patch tokens, the expectation and variance are computed across the spatial dimension, and $\epsilon = 10^{-6}$ for numerical stability.

Algorithm 1 Summary of key iREPA modifications to standard REPA training

```
# 1. Conv projection instead of MLP
proj_layer = nn.Conv2d(D_in, D_out, kernel_size=3, padding=1)

# 2. Spatial normalization on encoder features [B, T, D]
x = x - gamma * x.mean(dim=1, keepdim=True)
x = x / (x.std(dim=1, keepdim=True) + 1e-6)
```

4.1 EXPERIMENTS

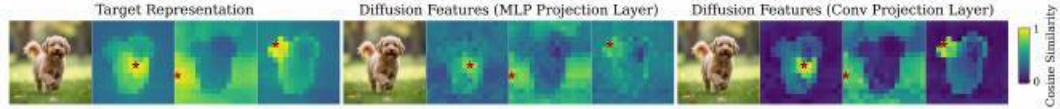
In this section, we quantify the improvement from the iREPA recipe through extensive experiments on Imagenet 256 × 256, always reporting iREPA against the matched REPA baseline. We organize the evaluation around the following questions:

- Can iREPA consistently improve the convergence speed of diffusion transformers over REPA across diverse external representations? (Figure 7, 8, 12, 13, 15)
- Is iREPA scalable in terms of model size and generalize across variations in training settings? (Table 1 [a,b,c], 2, 4, 5, 6, and, Figure 12, 16)
- Does iREPA generalize across more recent representation alignment methods such as REPA-E (Leng et al., 2025a), MeanFlow w/ REPA (Geng et al., 2025)? and pixel-space diffusion models such as JiT w/ REPA (Li & He, 2025b)? (Table 3 [a,b])

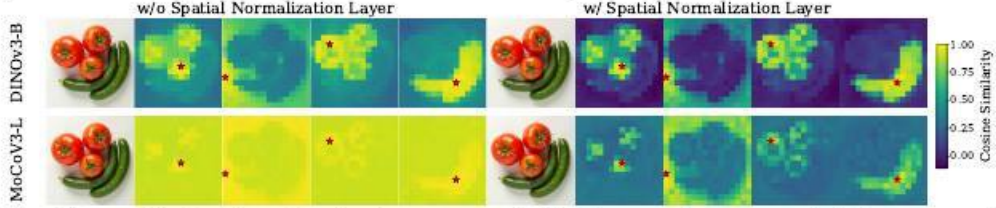
Convergence Speed. We track the FID-vs-iterations convergence of iREPA against REPA across six diverse vision encoders (DINOv3-B, WebSSL-1B, PE-Core-G, CLIP-L, MoCov3, PE-Lang-G) and two model sizes (SiT-XL/2 and SiT-B/2). As Fig. 7 shows, the iREPA curve sits below the REPA curve throughout training in every configuration—reaching any given FID target in fewer iterations—so the convergence-speed advantage holds uniformly across both the target-representation and model-size axes.

Target representation. We measure how the iREPA gain generalizes across vision encoders in Fig. 8 and Table 4. The improvement is universal: iREPA improves generation quality for *every* vision encoder tested, with no exceptions. The full comparison across all 27 encoders (Appendix C, E) shows the same one-directional pattern.

Encoder size. Table 1a reports the iREPA gain across three encoder sizes within the PE family: PE-B (90M), PE-L (320M), and PE-G (1.88B). iREPA improves generation at every encoder size—and, notably, the relative FID improvement grows monotonically with encoder scale, from 22.2% (PE-B)



(a) *Simpler projection layer for REPA.* Comparing projection choices on the same target: the standard MLP projection (middle) visibly washes out the spatial structure present in the target representation (left), whereas the simpler convolution layer (right) preserves it far more faithfully.



(b) *Spatial normalization layer.* Patch tokens of pretrained vision encoders carry a global component that suppresses spatial contrast—leaving tokens in one semantic region (*e.g.*, tomato) with substantial cosine similarity to unrelated tokens (*e.g.*, background or cucumber). Removing this global component sharpens the contrast between patch tokens, which is the property that predicts better generation.



(c) *Overall impact of improved training recipe (iREPA) on spatial structure of diffusion features.*

Figure 6: Motivating the two straightforward modifications (iREPA) that enhance spatial feature transfer from target representation to diffusion features. Panels (a) and (b) isolate where spatial information is lost (MLP projection, global component), and (c) shows the combined effect—iREPA restores markedly more of the target’s spatial structure in the diffusion features. All results reported with SiTL/2 with REPA at 100K.

Enc. Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$	Model Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$	Aln. Depth	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$
PE-B (90M)	59.6	22.5	6.23	0.63	0.60	SiT-B	27.5	49.50	7.00	0.46	0.59	Layer 4	28.1	49.0	6.83	0.46	0.60
+iREPA	73.3	17.5	6.04	0.66	0.62	+iREPA	34.1	43.37	7.87	0.50	0.60	+iREPA	41.7	36.0	6.61	0.52	0.62
PE-L (320M)	48.6	28.7	6.53	0.59	0.60	SiT-L	55.7	24.10	6.25	0.62	0.60	Layer 6	26.9	50.6	7.00	0.46	0.59
+iREPA	76.3	17.6	6.36	0.65	0.61	+iREPA	66.9	20.28	6.14	0.63	0.62	+iREPA	42.4	35.3	6.90	0.53	0.62
PE-G (1.88B)	42.7	32.3	6.70	0.57	0.59	SiT-XL	70.3	19.06	5.83	0.65	0.61	Layer 8	24.7	54.8	7.35	0.44	0.58
+iREPA	70.8	19.5	6.35	0.64	0.61	+iREPA	77.9	16.96	6.26	0.66	0.61	+iREPA	38.3	38.9	7.19	0.51	0.62

(a) Vision Encoder size

(b) Model size

(c) Encoder depth (SiT-B/2)

Table 1: **Variation in training settings.** Three orthogonal sweeps—(a) encoder size, (b) model size, (c) alignment depth—each pairing baseline REPA against +iREPA (highlighted). +iREPA lowers FID in *every* one of the nine pairs, with no regressions on the headline FID/IS metrics; the gain persists as we scale the encoder, scale the model, and vary the alignment depth. Unless otherwise specified, results use DINOv2 as encoder, SiT-XL/2, 100K steps, and vanilla-REPA as baseline.

to 38.8% (PE-L) to 39.6% (PE-G). The largest encoder, which fared worst under baseline REPA, benefits the most.

Scalability. Table 1b reports the iREPA gain across model scales (SiT-B, SiT-L, SiT-XL). The spatial improvements hold at every scale, and their benefit does not wash out as models grow—the relative gains are, if anything, larger for larger models—indicating that the spatial improvements scale favorably rather than saturating with model size.

Encoder depth. Table 1c reports the iREPA gain across alignment depths (SiT-B/2, 100K iterations, DINOv3-B as target). iREPA improves over baseline REPA at every alignment depth tested, so the benefit is not tied to a particular choice of alignment layer.

Abalation on different components. Table 2 isolates the contribution of each component. Both the spatial normalization layer and the convolutional projection layer individually improve generation quality over baseline REPA, the two are complementary, and combining them yields the best FID for every encoder—confirming that each modification contributes and that their gains add rather than overlap.

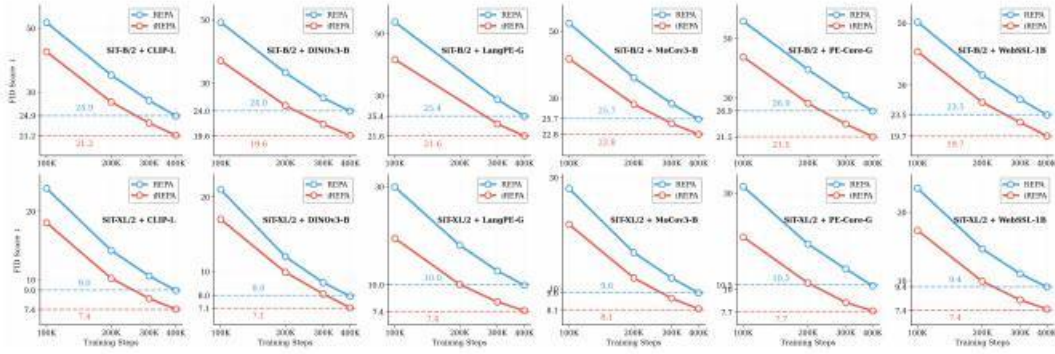


Figure 7: **Accentuating spatial features consistently improves convergence speed.** FID-vs-iterations curves for iREPA vs. REPA: across encoders and model sizes, the iREPA curve stays below the REPA curve throughout training, reaching each FID target in fewer iterations.

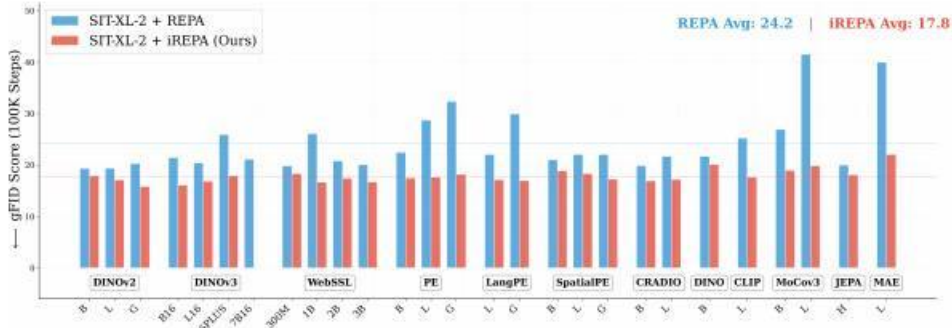


Figure 8: **Variation in target representation.** iREPA vs. REPA on all 27 vision encoders. The improvement is universal—iREPA lowers FID for every one of the 27 encoders, with no exceptions—so accentuating spatial transfer helps regardless of which target representation is used. See Appendix C for more results across diverse settings.

Training recipe. Finally, we test whether the gain is specific to vanilla REPA by porting the spatial improvements onto other recipes—REPA-E (Leng et al., 2025a) and MeanFlow w/ REPA (Geng et al., 2025). Table 3 shows the improvement transfers intact: iREPA delivers convergence gains on top of each recipe, so the benefit is a property of accentuating spatial transfer rather than of any single training pipeline.

Classifier-free guidance. Table 4 reports iREPA vs. REPA with CFG across vision encoders at 400K training iterations. The advantage is robust to the evaluation regime: iREPA converges faster both without and with classifier-free guidance, so the gain is not an artifact of a particular sampling setting.

Pixel-space diffusion (JiT). We further test iREPA on pixel-space diffusion with JiT-B (Li & He, 2025b). Figure 9 compares REPA and iREPA under JiT (Li & He, 2025b): accentuating spatial transfer again converges faster across every vision encoder considered (e.g., DiNov2, DiNov3, PE, etc.), showing the gain carries over from latent-space to pixel-space diffusion.

- **Finding 3.** Accentuating transfer of spatial structure from target representation to diffusion features helps boost convergence speed with representation alignment (REPA).

5 RELATED WORK

We discuss the most relevant related work here and provide a detailed discussion in Appendix H.

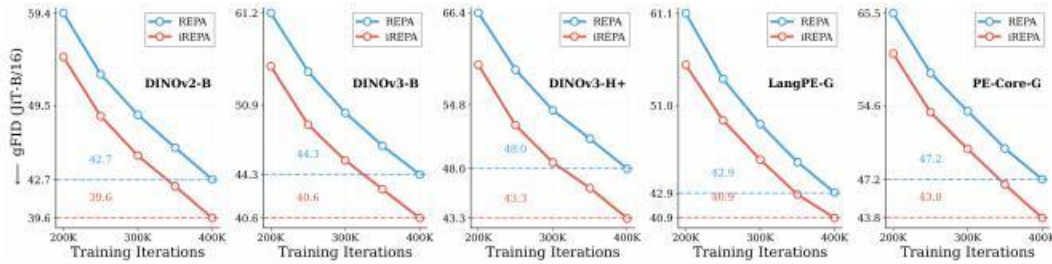


Figure 9: **Convergence comparison with pixel-space diffusion (JiT).** FID-vs-iterations curves for REPA vs. iREPA with pixel-space JiT (Li & He, 2025b) across vision encoders. The latent-space result carries over to pixel space: the iREPA curve sits below REPA throughout training for every encoder, reaching each FID target sooner. All results use JiTB/16 (Li & He, 2025b), batch size 256, and no classifier-free guidance. See Table 10 for further results.

Method	DINOv2-B			DINOv3-B			WebSSL-1B			PE-Core-G		
	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓
Baseline REPA	19.06	70.3	5.83	21.47	63.4	6.19	26.10	53.0	6.90	32.35	42.7	6.70
iREPA (w/o spatial norm)	18.52	73.3	6.11	17.76	74.7	5.81	21.17	64.6	6.27	24.97	57.4	6.21
iREPA (w/o conv proj)	17.66	72.8	6.03	18.28	70.8	6.18	18.44	71.0	6.22	21.72	61.5	6.26
iREPA (full)	16.96	77.9	6.26	16.26	78.8	6.14	16.66	77.5	6.18	18.19	75.0	6.03

Table 2: **Ablation on different components.** Each block adds components to baseline REPA (top row) and reports the effect per encoder at 100K steps on SiT-XL/2. Both the spatial normalization layer and the convolutional projection layer (§4) individually lower FID relative to REPA, and the full iREPA (both together, highlighted) attains the best FID and IS in all four encoders—showing the two modifications are complementary rather than redundant.

Representation alignment for generation. Many recent works explore the use of external representations for improving diffusion model training (Pernias et al., 2023; Fuest et al., 2024). Notably, (Yu et al., 2024; Yao & Wang, 2025; Leng et al., 2025a;b; Kouzelis et al., 2025) report large, measurable gains from aligning internal diffusion features with clean image features from a pretrained vision encoder, and (Zhang et al., 2025; Wu et al., 2025) extend the idea to video and 3D generation respectively, while (Ma et al., 2025a) applies it to unified models. Across these successes, however, the quantitative question of *which* property of the target representation produces the gain has gone unmeasured. This paper answers that question directly—identifying spatial structure as the predictive factor—and converts the answer into an improved training recipe.

Spatial vs global information tradeoff in pretrained vision encoders. Recent works study the tradeoff between global and spatial information in pretrained vision encoders (Bolya et al., 2025; Siméoni et al., 2025). (Siméoni et al., 2025) show that continued self-supervised training increases the similarity between the global CLS token and patch tokens, degrading dense spatial tasks, while (Bolya et al., 2025; Heinrich et al., 2024) explicitly train spatial-tuned models for such tasks. Our contribution is to quantify this tradeoff specifically for generation: across 27 encoders, spatial structure predicts generation quality far more strongly than global accuracy ($|r| > 0.85$ vs. $|r| = 0.26$). We hope this quantitative evidence motivates future work on selecting and training external representations for generation.

6 CONCLUSION

This paper asks, and answers with numbers, what truly drives the effectiveness of representation alignment: the global information or the spatial structure of the target representation? Our large-scale head-to-head analysis over 27 encoders and 3 model scales settles the question—spatial structure predicts generation quality far more strongly than global accuracy ($|r| > 0.85$ vs. $|r| = 0.26$, a $> 3\times$ gap that holds across model sizes), overturning the prevailing accuracy-first assumption. Acting on this, our two simple modifications (iREPA, <4 lines of code) that accentuate spatial transfer improve convergence speed over REPA in *every* tested setting—across vision encoders, model sizes, and training recipes. The size and consistency of these margins argue for revisiting the fundamental

Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
WebSSL-1B	52.8	26.5	5.20	0.620	0.585
+iREPA-E	87.0	13.2	5.28	0.699	0.598
PE-G	50.9	25.9	5.68	0.612	0.576
+iREPA-E	80.0	16.4	5.40	0.667	0.616
DINOv3-B	82.2	14.4	4.68	0.694	0.596
+iREPA-E	93.6	11.7	4.57	0.703	0.613
DINOv2-B	87.5	12.9	5.40	0.708	0.586
+iREPA-E	91.3	12.1	4.86	0.712	0.602

(a) REPA-E (SiT-XL/2)

Encoder	w/o CFG				w/ CFG			
	4 NFE		1 NFE		4 NFE		1 NFE	
	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$
WebSSL-1B	27.22	51.40	24.14	58.67	87.85	16.59	69.06	23.74
+iREPA	31.48	45.67	27.33	55.70	100.74	13.89	78.67	20.69
DINOv3-B	28.36	49.58	25.47	57.01	93.30	15.56	72.41	22.55
+iREPA	33.63	44.52	29.67	53.75	124.54	11.05	98.93	17.32
DINOv2-B	28.77	48.87	25.43	56.63	94.40	15.28	71.95	22.17
+iREPA	33.63	44.52	29.67	53.75	111.44	12.59	90.25	18.62

(b) MeanFlow w/ REPA (SiT-B/2)

Table 3: **Variation in training recipes.** The spatial improvements transfer to other recipes: (a) REPA-E and (b) MeanFlow w/ REPA, each pairing the base recipe against its +iREPA variant (highlighted). In (a), +iREPA-E lowers FID for all four encoders—by up to 13.3 points (WebSSL-1B: 26.5 $\rightarrow$ 13.2). In (b), +iREPA improves FID and IS at both 1 and 4 NFE, with and without CFG, across all encoders. For MeanFlow we report 1 and 4 NFEs with CFG 2.0; all results are at 100K training iterations.

Vision Encoder	Steps	w/o CFG					w/ CFG				
		IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	100K	69.20	19.3	5.89	0.64	0.61	157.2	6.35	5.91	0.769	0.536
+iREPA	100K	77.92	16.9	6.26	0.66	0.61	179.3	5.15	6.23	0.783	0.544
DINOv2-B	400K	127.4	7.76	5.06	0.70	0.66	263.0	1.98	4.60	0.799	0.610
+iREPA	400K	128.6	7.52	4.89	0.71	0.65	268.8	1.93	4.59	0.799	0.600
DINOv3-B	100K	63.64	21.4	6.14	0.63	0.60	144.0	7.57	6.09	0.762	0.526
+iREPA	100K	78.79	16.2	6.14	0.66	0.61	181.9	4.87	6.10	0.780	0.547
DINOv3-B	400K	126.7	8.10	5.06	0.70	0.66	261.2	1.99	4.58	0.799	0.609
+iREPA	400K	132.9	7.13	4.93	0.71	0.66	272.4	1.89	4.58	0.799	0.600
WebSSL-1B	100K	53.87	25.5	6.57	0.61	0.59	124.0	9.59	6.37	0.756	0.515
+iREPA	100K	77.47	16.6	6.18	0.66	0.61	177.6	5.09	6.11	0.787	0.538
WebSSL-1B	400K	116.9	9.39	5.14	0.70	0.64	250.5	2.24	4.61	0.809	0.580
+iREPA	400K	130.8	7.48	4.91	0.70	0.65	271.6	1.90	4.58	0.798	0.609
PE-Core-G	100K	42.74	32.3	6.70	0.57	0.59	97.2	14.1	6.56	0.714	0.525
+iREPA	100K	75.01	18.1	6.03	0.64	0.61	176.8	5.66	6.08	0.771	0.544
PE-Core-G	400K	109.4	10.4	5.00	0.69	0.64	238.4	2.44	4.57	0.805	0.585
+iREPA	400K	132.0	7.78	5.02	0.70	0.65	275.4	1.93	4.59	0.796	0.606

Table 4: **REPA vs. +iREPA across encoders, with and without classifier-free guidance.** Each encoder is shown at 100K and 400K steps (SiT-XL/2 base model), pairing REPA against +iREPA (highlighted). +iREPA is the better row on the headline IS/FID metrics in every pair and under both guidance settings—the gains are largest early in training (e.g., PE-Core-G at 100K: FID 32.3 $\rightarrow$ 18.1 w/o CFG) and persist at 400K, consistent with faster convergence to the same quality. See Appendix E for the full SiT-B/2 & SiT-L/2 breakdown across all encoders.

working mechanism of representational alignment and how to better leverage it for training generative models.

REPRODUCIBILITY STATEMENT

We provide all implementation details and hyperparameters in Appendix G. Our code, model checkpoints, and analysis results will be released upon acceptance.

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A DETAILED CORRELATION ANALYSIS

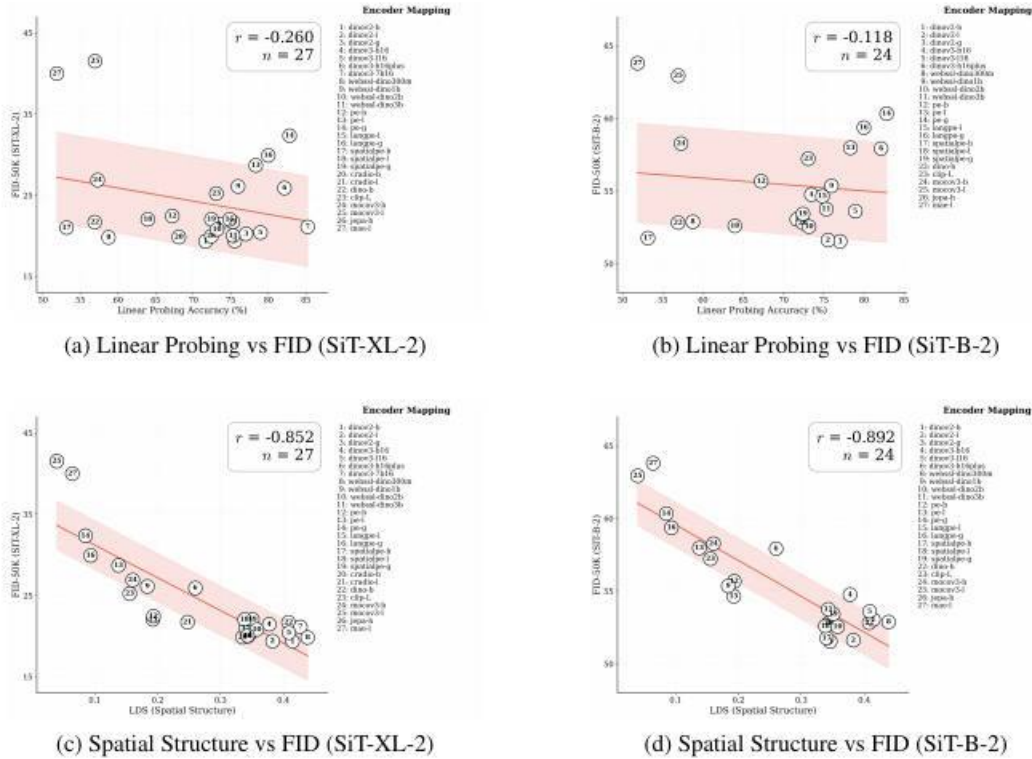


Figure 10: Detailed correlation analysis with encoder labels. The same head-to-head comparison as in the main text, now with every encoder labeled (see legend). Top row: linear probing accuracy vs. FID (SiT-XL-2 and SiT-B-2). Bottom row: spatial structure (LDS) vs. FID. Reading the labeled points, the spatial-structure metric orders the encoders in close agreement with FID at both scales, whereas linear probing does not—so its far stronger correlation is not driven by a few outliers.

B SPATIAL SELF-SIMILARITY METRICS

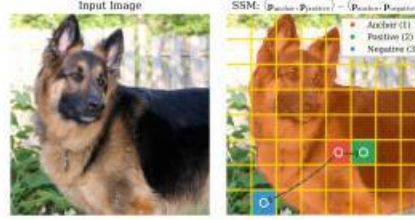


Figure 11: **Explaining semantic region self-similarity (SRSS) metric.** Visual explanation of SRSS metric: For each image, we compute a mask using SAM2 and select anchor-positive-negative triplets. The SSM measures the difference in cosine similarity between anchor-positive pairs (within mask) and anchor-negative pairs (outside mask). Intuitively, larger values indicate that patches on the same semantic region are more similar than patches on unrelated regions — indicating better spatial structure preservation.

Setup. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we extract patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ with $T = H \times W$ and indices $t \in [T]$ placed on an $H \times W$ lattice. Let $d : [T] \times [T] \rightarrow \mathbb{N}$ be the Manhattan distance, and let the (cosine) self-similarity kernel be

$$K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2} \in [-1, 1],$$

a standard local self-similarity measure (Shechtman & Irani, 2007).

- *Patch token representation:* $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ where $T = H \times W$ is the spatial grid of patches (patch index set $[T]$). Also let $d : [T] \times [T] \rightarrow \mathbb{N}$ be Manhattan distance on the $H \times W$ grid.
- *Self-similarity kernel:* $K : [T] \times [T] \rightarrow \mathbb{R}$ measuring self-similarity (Shechtman & Irani, 2007) between patch tokens. We use the cosine kernel $K_{\mathcal{X}}(t, t') = \langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle / (\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2)$.
- *Spatial self-similarity metric:* a functional $m : \mathcal{K} \times \mathcal{D} \rightarrow \mathbb{R}$ measures how self-similarity K between patch tokens varies with lattice distance d . Intuitively, larger values indicate stronger spatial organization (near patches more similar to each other than far away patches).

We next discuss several straightforward metrics for measuring the spatial self-similarity structure of the target representations.

Local vs. Distant Similarity (LDS). We first consider a simple correlogram contrast (local vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') < r_{\text{near}}] - \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') \geq r_{\text{far}}].$$

where r_{near} and r_{far} are the hyperparameters. By default we use $r_{\text{near}} = r_{\text{far}} = H/2$. We found correlation to be robust to their exact choices. Intuitively, larger values indicate that *on average*, patches that are closer to each other are more similar than patches that are further away.

Correlation Decay Slope (CDS). Given the patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$, we compute the spatial correlogram $g_{\mathcal{X}}(\delta) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') = \delta]$ for distances $\delta \in \{1, \dots, \Delta\}$. We then fit a least-squares line $\hat{g}_{\mathcal{X}}(\delta) \approx \alpha + \beta\delta$ and define:

$$\text{CDS}(\mathcal{X}) = -\hat{\beta}$$

where $\hat{\beta}$ is the fitted slope. Larger CDS values indicate faster similarity decay with distance, suggesting stronger spatial organization.

Semantic-Region Self-Similarity (SRSS). For each image, we first sample a binary mask $\mathbf{M} \in \mathbb{R}^{H \times W}$ containing a random object using SAM2 (Ravi et al., 2024) and downsampled to $H \times W$. We then select three types of points: anchor, positive, and negative. The anchor and positive points are sampled from within the mask (should have similar semantics), while the negative point is sampled from outside the mask (less related to anchor). Conceptually, if the encoder feature representation preserves the spatial structure, the anchor and positive points should have higher cosine similarity,

while the anchor and negative points should have lower cosine similarity. Thus, we define the Spatial Structure Metric (SSM) as:

$$\text{SSM}(\mathbf{P}) = \mathbb{E}_{\text{anchor} \in \mathbf{M}} [\cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{pos}}) - \cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{neg}})]$$

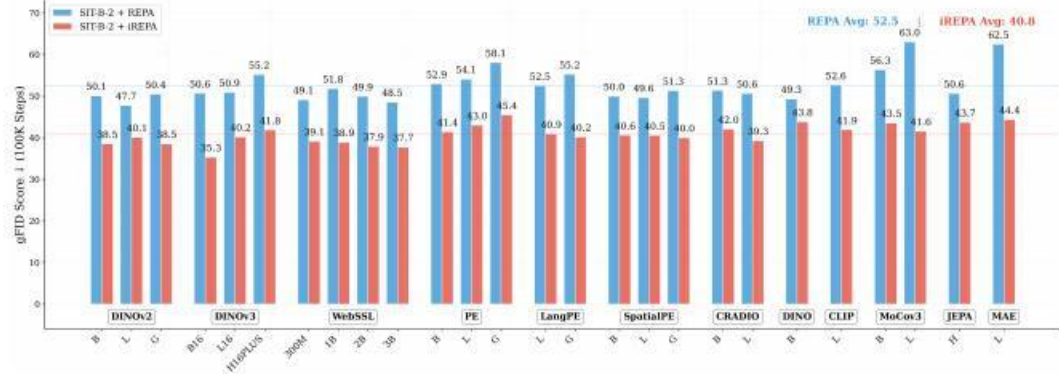
where positive points are within Manhattan distance $d \leq r_{\text{near}}$ from the anchor within the mask, and negative points are at distance $d \geq r_{\text{far}}$ outside the mask.

RMS Spatial Contrast (RMSC). Finally, we consider a simple contrast metric to capture the spatial contrast between patch token representations. Given normalized features $\hat{\mathbf{x}}_t = \mathbf{x}_t / \|\mathbf{x}_t\|_2$ for each patch t , we compute:

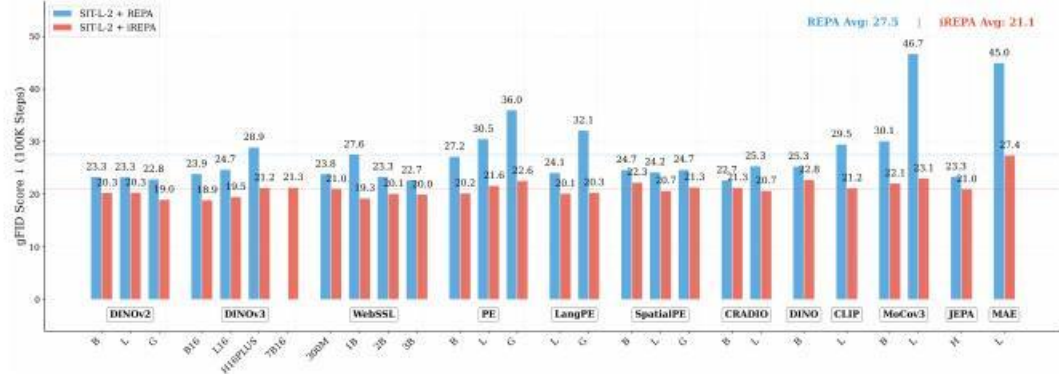
$$\text{RMSC}(\mathcal{X}) = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{x}}_t - \bar{\mathbf{x}}\|_2^2}$$

where $\bar{\mathbf{x}} = \frac{1}{T} \sum_{t=1}^T \hat{\mathbf{x}}_t$ is the mean of normalized features across all spatial locations. Higher RMSC values indicate greater spatial diversity in the feature representations, suggesting preserved spatial structure, whereas lower values indicate more uniform feature distributions indicating a loss of spatial structure (typically happens global information diminishes spatial structure (Siméoni et al., 2025)).

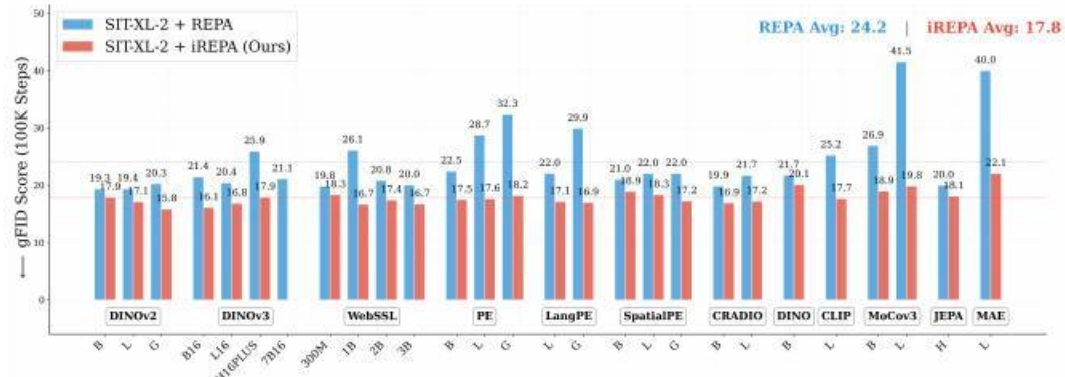
C VISION ENCODER VARIATION



(a) SiT-B-2 (130M parameters)



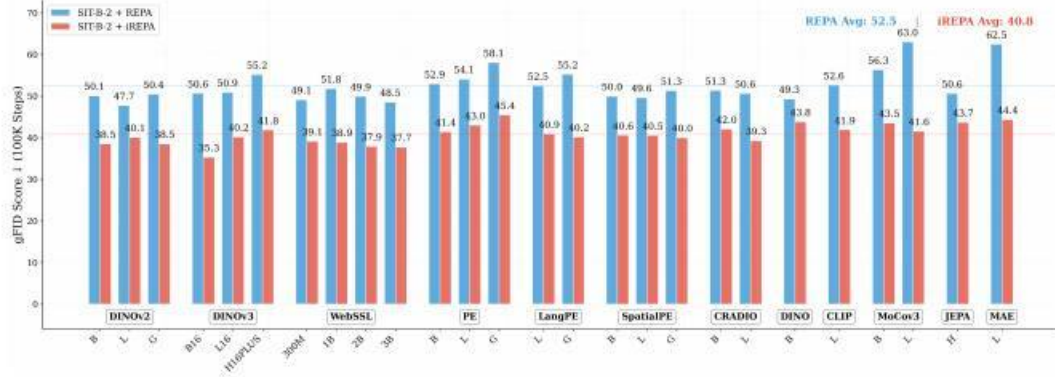
(b) SiT-L-2 (458M parameters)



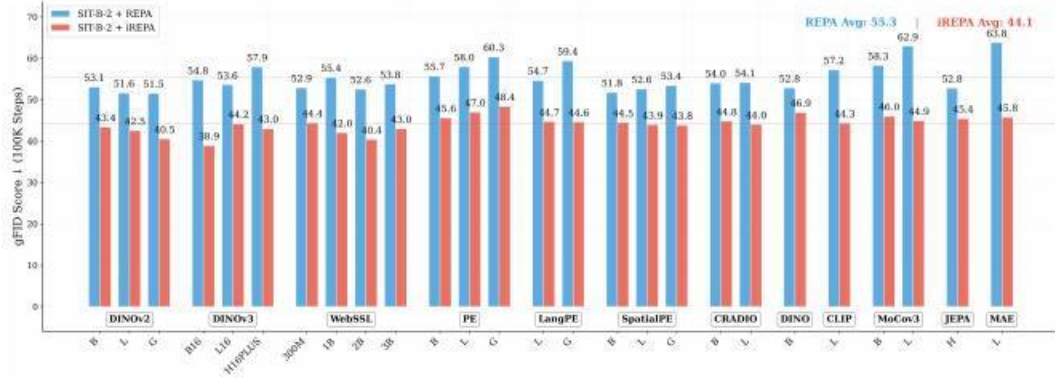
(c) SiT-XL-2 (675M parameters)

Figure 12: **Variation in vision encoders.** Per-encoder FID for iREPA vs. REPA at three model sizes (SiT-B-2, SiT-L-2, SiT-XL-2). iREPA improves generation quality for every encoder at every model size—the gain holds across the full encoder×model-size grid without exception.

D ENCODER DEPTH VARIATION



(a) SiT-B-2 with 6 encoder layers



(b) SiT-B-2 with 8 encoder layers

Figure 13: **Effect of encoder depth on iREPA performance.** Per-encoder FID for iREPA vs. REPA on SiT-B-2 at two alignment depths (6 vs. 8 layers). iREPA improves over REPA for every encoder at both depths—so the gain is robust to the alignment-depth choice—with the 8-layer setting yielding a slightly larger improvement.

E COMPREHENSIVE RESULTS ACROSS VISION ENCODERS

E.1 SiT-B/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	59.44	24.84	6.46	0.584	0.645
+iREPA	69.70	21.26	6.64	0.601	0.644
DINOv2-B	64.92	22.75	6.54	0.593	0.653
+iREPA	70.88	21.40	6.77	0.597	0.653
DINOv2-L	66.65	22.46	6.53	0.594	0.650
+iREPA	72.41	20.99	6.84	0.600	0.650
DINOv3-B16	62.57	23.91	6.77	0.584	0.647
+iREPA	75.48	19.65	6.68	0.606	0.649
DINOv3-H16+	61.91	24.82	6.57	0.579	0.645
+iREPA	71.76	21.80	6.84	0.592	0.655
DINO-B	58.37	24.71	6.32	0.586	0.635
+iREPA	57.55	24.41	6.43	0.587	0.638
LangPE-G	58.42	25.37	6.32	0.583	0.641
+iREPA	66.88	21.62	6.65	0.604	0.649
MoCov3-B	57.61	25.60	6.25	0.579	0.640
+iREPA	61.41	22.80	6.46	0.596	0.639
PE-B	58.86	25.78	6.44	0.58	0.64
+iREPA	69.50	21.15	6.66	0.601	0.659
PE-G	56.09	26.84	6.45	0.568	0.641
+iREPA	70.43	21.52	6.58	0.598	0.651
SpatialPE-B	59.19	24.28	6.37	0.586	0.646
+iREPA	61.50	22.71	6.41	0.604	0.644
SpatialPE-L	62.13	23.87	6.51	0.59	0.65
+iREPA	68.83	21.07	6.59	0.607	0.651
WebSSL-1B	63.20	23.48	6.33	0.593	0.646
+iREPA	73.54	19.74	6.63	0.612	0.646
WebSSL-2B	66.56	22.27	6.69	0.596	0.654
+iREPA	72.13	20.47	6.89	0.609	0.648
WebSSL-300M	62.06	23.64	6.48	0.590	0.646
+iREPA	66.83	21.46	6.71	0.606	0.653
WebSSL-3B	65.91	22.55	6.53	0.598	0.654
+iREPA	71.98	20.64	6.71	0.607	0.650

Table 5: SiT-B/2 across vision encoders at 400K iterations: REPA vs. +iREPA (highlighted). Reading down each encoder pair, +iREPA is the better row on FID and IS in all 16 encoders (a 16/16 sweep with no regressions on the headline metrics). All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

E.2 SiT-L/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	107.7	10.6	5.20	0.682	0.643
+iREPA	115.8	9.41	5.15	0.689	0.653
DINOv2-B	113.3	9.61	5.11	0.684	0.653
+iREPA	113.9	9.36	5.11	0.690	0.655
DINOv3-B16	115.3	9.41	5.24	0.687	0.651
+iREPA	118.1	9.06	5.25	0.685	0.650
DINOv3-H16+	110.7	10.6	5.32	0.677	0.652
+iREPA	120.5	9.64	5.29	0.681	0.663
LangPE-G	102.8	11.2	5.13	0.684	0.637
+iREPA	112.7	9.37	5.02	0.690	0.653
MoCov3-B	102.1	11.0	5.14	0.685	0.635
+iREPA	104.7	10.1	5.00	0.690	0.641
PE-G	98.38	12.6	5.31	0.670	0.646
+iREPA	117.1	9.65	5.33	0.683	0.651
WebSSL-1B	107.9	10.5	5.13	0.686	0.644
+iREPA	121.5	8.62	5.01	0.696	0.653

Table 6: SiT-L/2 across vision encoders at 400K iterations: REPA vs. +iREPA (highlighted). +iREPA is the better row on FID and IS for every one of the 8 encoders, with the largest FID gains on the encoders REPA struggled with most (e.g., PE-G: 12.6 $\rightarrow$ 9.65). All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

F ADDITIONAL EXPERIMENTAL RESULTS

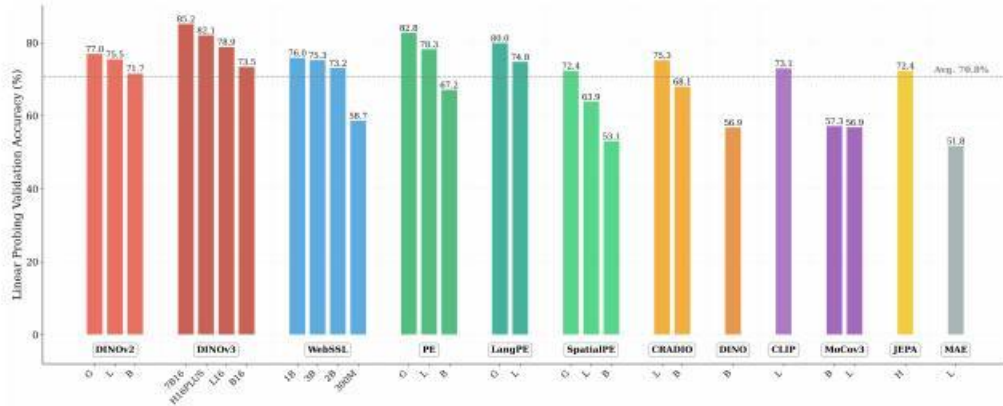


Figure 14: **Global information in mean patch tokens.** Beyond the [CLS] token, the *mean* of patch tokens itself linearly probes to high accuracy—direct evidence that it too carries substantial global semantic information. This global component boosts global performance but flattens the contrast between individual patch tokens, working against spatial-structure transfer. Removing it (subtracting the mean of patch tokens) is therefore a targeted way to strengthen spatial-structure transfer during representation alignment (§4).

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	262.97	1.98	4.60	0.799	0.610
+iREPA (Ours)	268.79	1.93	4.59	0.799	0.600
DINOv3-B	261.18	1.99	4.58	0.799	0.609
+iREPA (Ours)	272.41	1.89	4.58	0.799	0.600
WebSSL-1B	250.53	2.24	4.61	0.809	0.580
+iREPA (Ours)	271.59	1.90	4.58	0.798	0.609
PE-G	238.37	2.44	4.57	0.805	0.585
+iREPA (Ours)	275.36	1.93	4.59	0.796	0.606

Table 7: **Generation quality with classifier-free guidance.** Paired comparison of REPA against +iREPA (highlighted) with CFG (scale 2.0), all under matched SiT-XL/2 training for 400K iterations with 250 sampling steps. In every encoder pair, +iREPA is the better row on both IS and FID—so the improvement is 4/4 across encoders even in the strong-CFG regime.

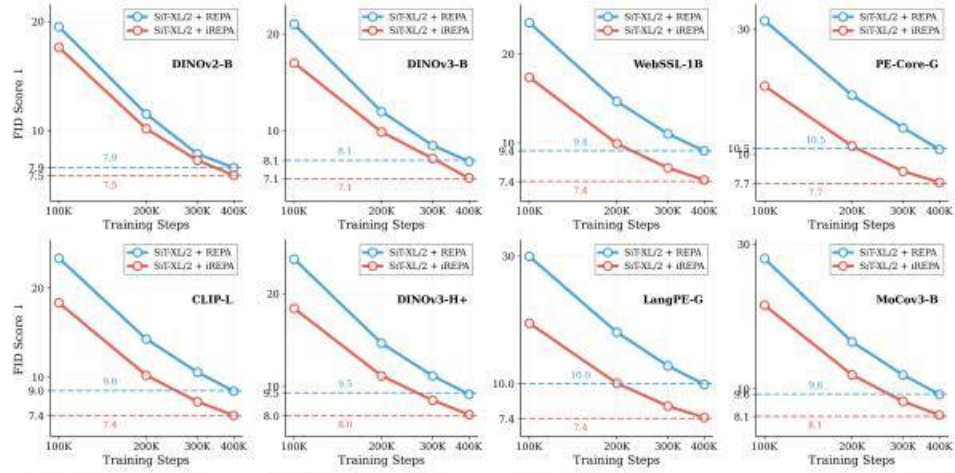


Figure 15: Accentuating spatial features helps consistently improve convergence speed. Results for SiT-XL/2 with REPA and iREPA.

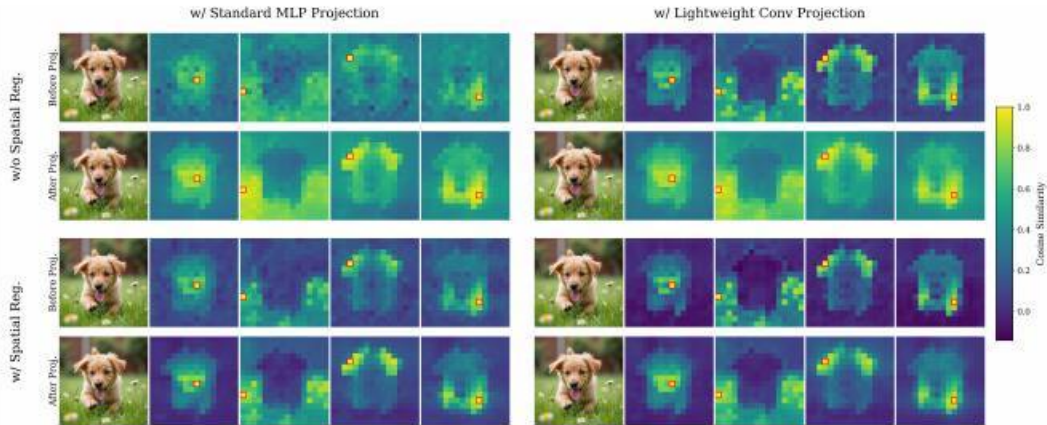


Figure 16: Visualizing the impact of two straightforward improvements to enhance spatial feature transfer. First, comparing projection choices side by side, the standard MLP projection (top-left) discards spatial information when transferring features from the pretrained encoder (after projection) to the diffusion model (before projection), while the simpler convolution layer (top-right) preserves that transfer far better. Second, vision encoder features often have limited spatial contrast (§2), leaving tokens in one semantic region (e.g., dog) with substantial cosine similarity to unrelated tokens (e.g., background); a simple spatial regularization layer sharpens this contrast and improves generation. Combining both modifications is best—it attains the lowest FID for every encoder (Table 2).

G IMPLEMENTATION DETAILS

Training setup. For a fair comparison, we hold the training setup identical to REPA (Yu et al., 2024) unless stated otherwise—so any measured iREPA gain is attributable to the two spatial modifications rather than to setup differences. All training uses the ImageNet (Deng et al., 2009) training split. For preprocessing, we adopt the ADM (Dhariwal & Nichol, 2021) protocol, center-cropping and resizing each image to 256×256 resolution. We use `stabilityai/sd-vae-ft-mse` VAE (AI, n.d.) throughout our diffusion training and inference. For spatial normalization layer we use $\gamma \in [0.6, 0.8]$ and for projection layer we use a convolutional layer with kernel size 3 and padding 1. For optimization, we use AdamW (Kingma & Ba, 2014; Loshchilov, 2017) with a constant learning rate of 1×10^{-4} , and a global batch size of 256. During training, we use `bfloat16` mixed precision and `torch.compile` to accelerate training, and gradient clipping and exponential moving average (EMA) to the generative model for stable optimization.

For REPA-E (Leng et al., 2025a) and JiT (Li & He, 2025a) experiments, we use the official open-source implementation. For MeanFlow (Geng et al., 2025) experiments, we received the implementation (including REPA) from original authors and adapt it to introduce the two straightforward changes for iREPA (§4).

Evaluation. We evaluate generation strictly under the ADM setup (Dhariwal & Nichol, 2021), so all reported numbers are directly comparable. Quality is measured on 50K generated images with Fréchet inception distance (gFID) (Heusel et al., 2017), structural FID (sFID) (Nash et al., 2021), inception score (IS) (Salimans et al., 2016), and precision/recall (Prec./Rec.) (Kynkäänniemi et al., 2019). Sampling follows REPA (Yu et al., 2024) using the SDE Euler-Maruyama sampler with 250 steps; for JiT (Li & He, 2025a) we use 50 inference steps per the official implementation.

H MORE DISCUSSION ON RELATED WORK

Classical spatial features in computer vision. Spatial feature extraction has long been fundamental to computer vision. Classical methods like SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), SURF (Bay et al., 2006), and ORB (Rublee et al., 2011) providing robust local descriptors for tasks ranging from object detection to image matching. While these handcrafted features excel at capturing local spatial patterns and geometric invariances, recent work in generative modeling has primarily focused on representations from modern self-supervised methods that demonstrate strong global classification performance, such as DINOv2 (Oquab et al., 2024) and CLIP (Radford et al., 2021). Our quantitative findings suggest a different perspective: because spatial structure is what predicts generation quality (§3), even classical spatial descriptors improve diffusion training once properly aligned—as we confirm empirically for SIFT, HOG, and VGG. This points to leveraging the full spectrum of spatial feature extractors, from handcrafted descriptors to modern learned representations, so long as they maintain strong spatial coherence.

Pretrained visual encoders for generative models. Pretrained visual encoders have supported generative models in several capacities: as discriminators to accelerate GAN convergence (Goodfellow et al., 2020; Sauer et al., 2021; 2022; 2023; Radford et al., 2021), as teachers in adversarial distillation for diffusion models (Sauer et al., 2024; Kang et al., 2024), and more recently as alignment targets. In GANs (Goodfellow et al., 2020), pretrained features have not only improved convergence speed but also enabled scaling to large datasets, as demonstrated by StyleGAN-XL (Sauer et al., 2022) and StyleGAN-T (Sauer et al., 2023) with CLIP (Radford et al., 2021) features. For diffusion models, adversarial distillation leverages pretrained encoders to guide student networks toward higher-fidelity samples, showing clear improvements in perceptual quality. In particular, REPA (Yu et al., 2024) aligns diffusion features with external encoders, demonstrating that representation alignment can improve both generation convergence and quality. Building on this direction, we focus not only on alignment but specifically on spatial structure perseverance rather than their discriminative capabilities.

Denosing transformers. Transformer architectures have become the dominant backbone for scalable generative modeling, with various formulations including diffusion transformers (Peebles & Xie, 2023) and flow matching variants (Ma et al., 2024). Recent architectures like GenTron (Chen et al., 2023) scale transformers to over 3B parameters for text-to-image synthesis. U-ViT (Bao et al., 2023) demonstrates that pure transformer backbones without U-Net inductive biases can achieve competitive

performance. ARDiT (Tang et al., 2024) and DART (Jing et al., 2024) explore autoregressive formulations that combine denoising with sequential generation, enabling flexible trade-offs between quality and speed. Despite these architectural advances, training these models from scratch remains computationally expensive, often requiring millions of iterations to achieve good generation quality. While representation alignment methods like REPA (Yu et al., 2024) demonstrate that pretrained features can dramatically accelerate convergence, the mechanism behind the improvement remained unquantified. Our head-to-head analysis pins it down: the benefit comes from preserving spatial structure ($|r| > 0.85$ with FID) rather than semantic alignment ($|r| = 0.26$), which explains why certain encoders accelerate training far more than others and guides the design of more effective alignment strategies. While (Wang et al., 2025c) propose improvements to REPA training through early stopping, they hypothesize that “REPA predominantly distills global semantic information while leaving structural information untapped.” In contrast, our analysis reveals that spatial structure (not global semantic information) already plays a very significant role in the effectiveness of REPA (§2, §3). Thus, while we indeed find that spatial structure remains underexploited (§4) in REPA, we surprisingly find that majority of improvements in REPA are already coming from introducing an inductive bias for spatial structure (not global information).

Denoising as self-supervised learning task. The connection between denoising and representation learning has been explored from multiple perspectives. Early work by Abstreiter et al. (2021) extended diffusion objectives for representation learning, demonstrating that denoising naturally learns meaningful features. SODA (Hudson et al., 2024) introduces a diffusion model with an information bottleneck to learn compact representations. Chen et al. (2024) deconstruct diffusion models and find that a simple denoising autoencoder suffices for strong self-supervised performance. Wang & He (2025) introduce a dispersive loss to encourage internal representation diversity in diffusion models, improving generative performance without external encoders. Similarly, Jiang et al. (2025) propose Self-Representation Alignment (SRA) to align a diffusion transformer’s latent features across noise levels, providing self-guidance without an auxiliary model. These works establish denoising as a fundamental self-supervised task that naturally encourages learning of robust features. Our findings complement this view by showing that when diffusion models are aligned with strong spatial representations from self-supervised encoders, both the generative and discriminative capabilities improve, suggesting that spatial structure preservation is a key factor in this synergy.

Scaling self-supervised vision encoders. Recent years have seen remarkable progress in scaling self-supervised vision models to unprecedented sizes and datasets. DINOv3 (Siméoni et al., 2025) trains a 7B parameter ViT on 1.7 billion images without labels by aligning representations from different augmentations. WebSSL (Fan et al., 2025) demonstrates that visual models trained on more than 2 billion images can match language-supervised models like CLIP (Radford et al., 2021) on vision-language tasks without language supervision. C-RADIO (Heinrich et al., 2024) combines multiple teacher models through distillation to create versatile encoders that excel across diverse visual domains. I-JEPA (Assran et al., 2023) explores predictive architectures that learn by predicting masked regions in representation space, instead of reconstructing pixels directly. SAM (Ravi et al., 2024) specializes in promptable segmentation through large-scale supervised training. The Perception Encoder family (Bolya et al., 2025; Cho et al., 2025) shows that intermediate features often outperform final representations for dense prediction tasks. These models are typically ranked on global tasks like image classification, but our results show that ranking does not carry over to generation: strong discriminative-benchmark performance does not translate into better generation quality, and across 27 encoders classification accuracy is only weakly predictive of it ($|r| = 0.26$). What predicts generation is spatial structure, regardless of classification accuracy. This argues that the evaluation metrics for self-supervised encoders should be reconsidered for generative applications, with spatial coherence weighted at least as heavily as semantic understanding.

I NOTE ON LLM USAGE

We use GPT-5 (OpenAI, 2025) for considering different variations of the spatial structure metrics discussed in the paper §3. Furthermore, all figures in the paper are directly generated from our experiment logs and checkpoints using Claude-Code (Anthropic, 2025). Additionally we use LLM help for searching and formulating relevant work in Appendix H. We use cursor in some parts to help with paper writing.

J ADDITIONAL DISCUSSION ON SPATIAL STRUCTURE METRICS

Spatial structure metrics can be used to measure the representation gap. Yu et al. (2024) measure the representation gap with linear probing, citing the increase in validation accuracy as evidence for REPA’s effectiveness. We show the spatial-structure metric (SSM) serves the same diagnostic role—and is better aligned with generation. As Figure 17 shows, representation alignment measurably narrows the SSM gap between a pretrained encoder (e.g., DINOv2) and the diffusion features, and iREPA narrows it further, tracking the observed generation gains.

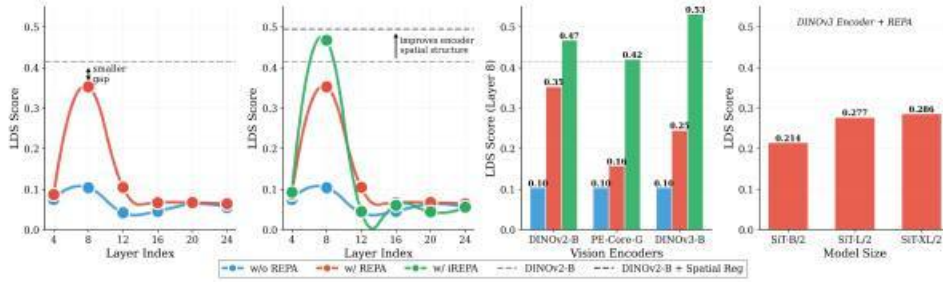


Figure 17: Representation alignment as bridging the spatial feature gap. **a)** Measured on the spatial-structure metric, representation alignment narrows the spatial feature gap between diffusion and vision-encoder patch features. **b)** iREPA (§4) closes more of this gap by accentuating the encoder’s spatial features (trading away some global information), which translates into better generation. **c)** the resulting improvement holds across every vision encoder tested. **d)** the spatial-structure gains do not saturate—they grow with model size. All results are reported with SiT-XL/2 at 400k iterations.

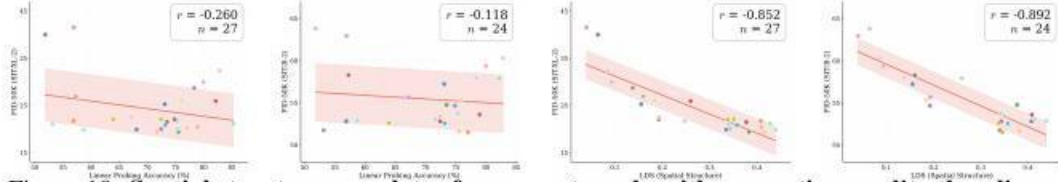


Figure 18: Spatial structure correlates far more strongly with generation quality than linear probing. Head-to-head correlation with FID across 27 vision encoders at two model scales. **Left two:** linear probing accuracy vs. FID is weak—SiT-XL-2 (Pearson $r = -0.26$) and SiT-B-2 ($r = -0.12$). **Right two:** spatial structure (LDS) vs. FID is strong—SiT-XL-2 ($r = -0.85$) and SiT-B-2 ($r = -0.89$). The spatial predictor is roughly 3–7 $\times$ stronger, and wins at both scales. See Figure 10 in Appendix for detailed plots with encoder labels.

K ADDITIONAL ANECDOTAL COMPARISONS

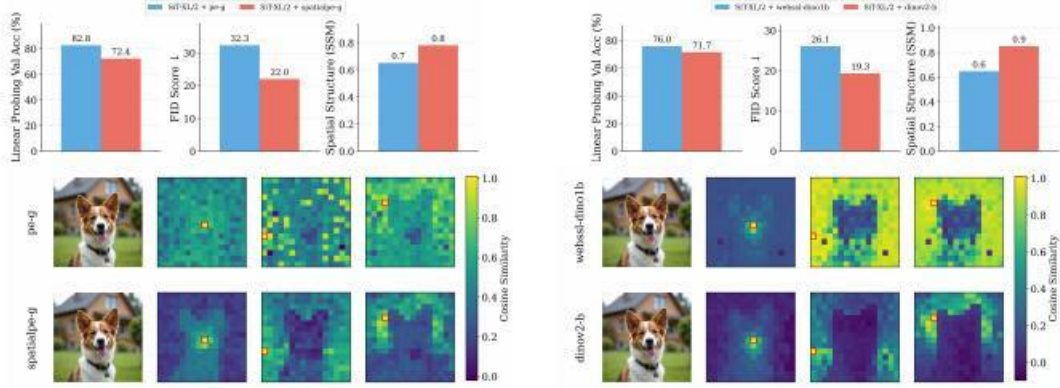


Figure 19: **Motivating anecdotes from recent SSL encoders — spatial structure matters.** **Top row:** two head-to-head pairs, each showing the accuracy leader losing on generation. **Left:** PE-g leads on accuracy (82.8% vs. 72.4%) but trails on FID (32.3 vs. 22.0) against Spatial-PE-g, and its spatial structure is 4× lower (LDS: 0.1 vs. 0.4). **Right:** WebSSL-dino1b leads on accuracy (76.0% vs. 71.7%) but trails on FID (26.1 vs. 19.3) against DINOv2-b, again with weaker spatial structure (LDS: 0.2 vs. 0.4). **Bottom row:** the spatial cosine-similarity maps make the pattern visible—the better-generating encoders (Spatial-PE-g, DINOv2-b) keep clear spatial coherence, while the classification-tuned ones (PE-g, WebSSL) lose it. In every case, LDS orders the pair the way FID does.

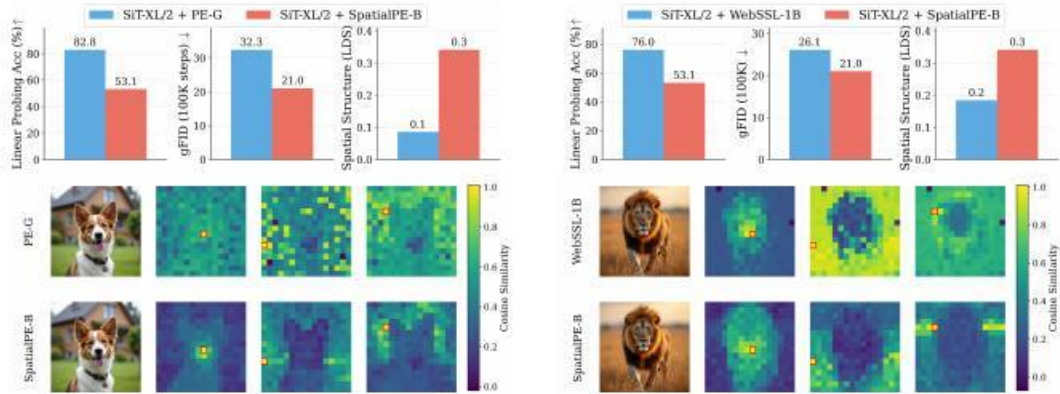


Figure 20: **Motivating examples — spatial structure matters.** **Top:** two head-to-head pairs exhibiting the same inverse relationship between ImageNet accuracy and generation quality. **Left:** PE_{core}-G, despite a 29.7-point accuracy lead (82.8% vs. 53.1%), generates worse than PE_{Spatial}-B (Bolya et al., 2025). **Right:** likewise, WebSSL-1B (Fan et al., 2025) leads on global accuracy (76.0% vs. 53.1%) yet generates worse. **Bottom:** spatial structure orders each pair the way generation does—the reliable predictor that global accuracy is not. See §3 for the spatial structure metric. All results reported at 100K using SiT-XL/2 and REPA.

L FURTHER DISCUSSION

Role of Spatial Normalization We visualize what spatial normalization does to vision-encoder representations. Many encoders carry a strong global component that suppresses the contrast between spatial tokens: tokens in one semantic region end up highly correlated with tokens in unrelated regions (e.g., background), blurring the spatial distinctiveness of the features. Spatial normalization removes this global overlay, raising the contrast between distinct spatial tokens—preserving more local spatial structure while cutting the dominance of the global information that works against generation quality.

The figures below make this effect concrete across several examples, showing side by side how normalization reshapes the feature similarity maps toward sharper spatial structure:

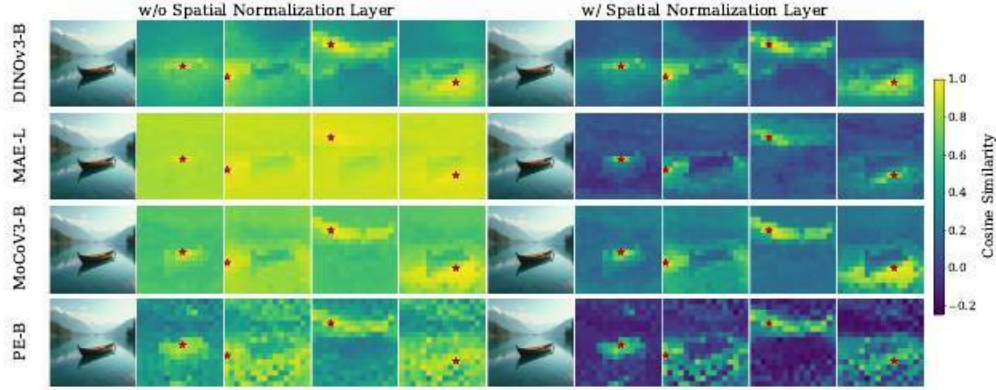


Figure 21: Visualizing the impact of spatial normalization (Example 1). Token-similarity heatmaps, side by side, before (left) and after (right) spatial normalization. Before, the global component induces high similarity across unrelated regions; after, the same tokens show sharper local structure and cleaner semantic boundaries—the qualitative signature of the spatial-contrast gain.

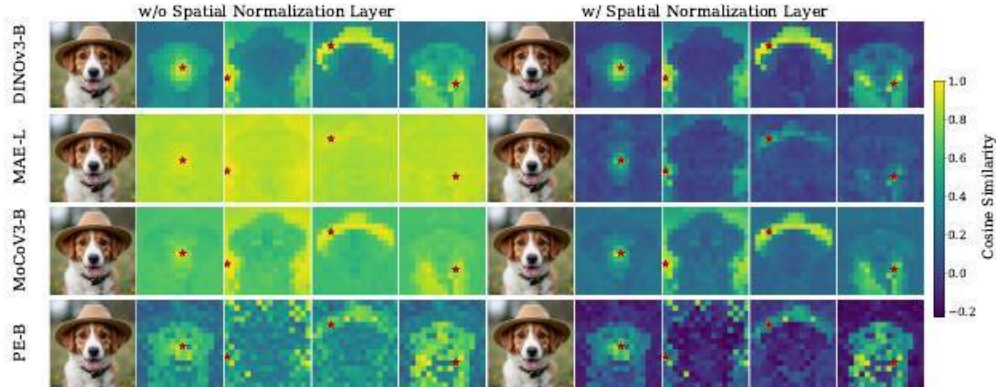


Figure 22: Visualizing the impact of spatial normalization (Example 3). Token-similarity heatmaps before (left) and after (right) spatial normalization. Before, the global component drives high similarity across unrelated regions; after, local structure is sharper and semantic boundaries more distinct—the same before/after contrast gain seen in the other examples.

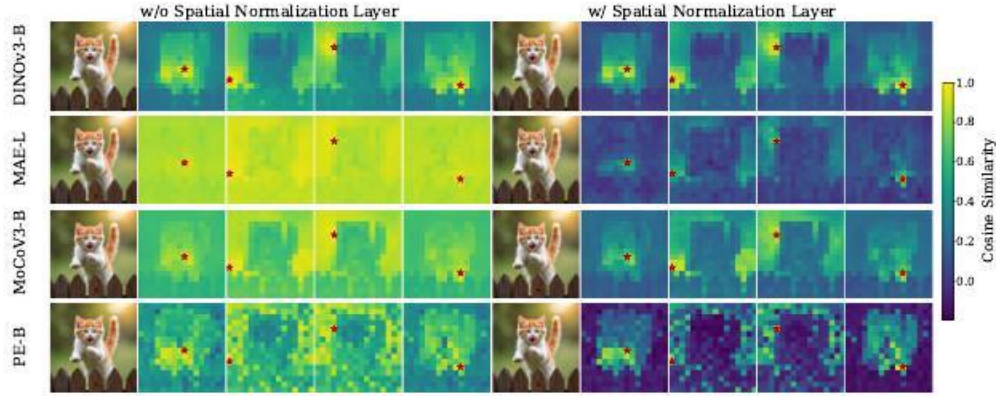


Figure 23: Visualizing the impact of spatial normalization (Example 2). Feature-similarity maps before (left) and after (right). The original features carry a global overlay that blurs the foreground/background distinction; after normalization the same tokens are more locally coherent, with clearer separation between semantic regions—again showing the contrast gain visually.

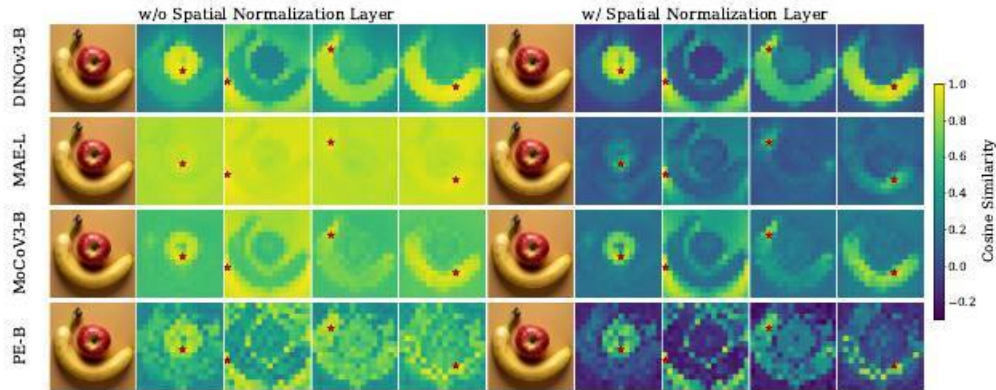


Figure 24: **Visualizing the impact of spatial normalization** (Example 4). Token-similarity heatmaps before (left) and after (right) spatial normalization. Before, the global component creates high similarity across unrelated regions; after, local structure is enhanced and semantic boundaries are more distinct—consistent with the before/after contrast gain across all four examples.

M ADDITIONAL RESULTS

M.1 ADDITIONAL RESULTS AT HIGHER RESOLUTIONS

Setup. We test whether the spatial improvements survive a change of resolution, running ImageNet-512 (Deng et al., 2009) under the same REPA (Yu et al., 2024) setup across pretrained encoders (DINOv2, DINOv3, WebSSL, PE, etc.), with SiT-XL and SiT-B at 50 NFE and no classifier-free guidance. Fig. 25 pairs REPA against +iREPA: the faster-convergence advantage carries over to 512×512 intact, and holds for every encoder tested—so the gain is not specific to the 256×256 resolution.

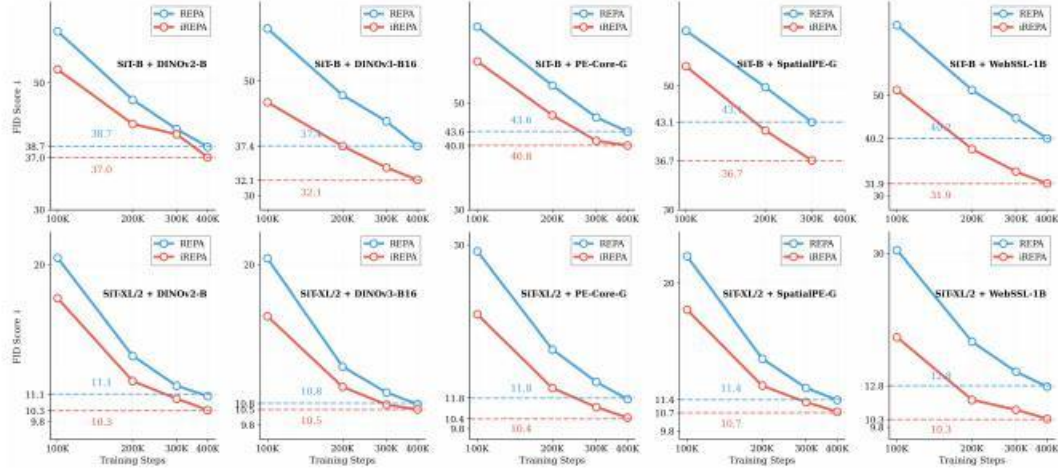


Figure 25: **Convergence results at 512x512 resolution.** iREPA vs. REPA: accentuating spatial features improves convergence speed at both resolutions—ImageNet-256 (Figure 7) and ImageNet-512 (above)—with the iREPA curve below REPA for every encoder here as well.

M.2 ADDITIONAL RESULTS ON MULTIMODAL T2I TASKS

Setup. To test generalization beyond ImageNet class-conditional generation, we move to multimodal text-to-image (T2I). Following REPA (Yu et al., 2024), we use MMDiT (Esser et al., 2024) as the diffusion backbone and apply REPA with various pretrained vision encoders (DINOv2, CLIP, WebSSL, PE, etc.). Matching REPA (Yu et al., 2024) exactly, we train all models for 150K steps at batch size 256 and evaluate with an ODE sampler at 50 NFE, so the comparison isolates the effect of the spatial modifications. Fig. 26 pairs REPA against +iREPA: the improvement transfers to the T2I setting and holds across every encoder—so the spatial-transfer benefit is not tied to the class-conditional ImageNet task.

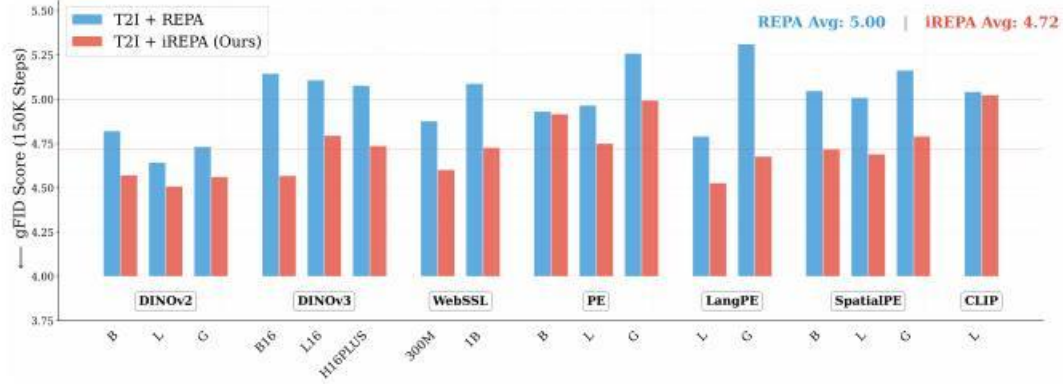


Figure 26: **Text-to-image generation across encoder variants.** iREPA vs. REPA: accentuating spatial-feature transfer improves convergence speed on both ImageNet (§4) and multimodal T2I (above), with the gain holding for every pretrained encoder tested (DINOv2, CLIP, WebSSL, PE, etc.).

M.3 CORRELATION ANALYSIS WITHOUT OUTLIERS (MoCOV3-L AND MAE-L)

As a robustness check, we repeat the Section 3 correlation analysis after removing the two identified outliers (MoCOV3-L and MAE-L). Figure 28 shows the updated correlations across model sizes: spatial structure retains its large predictive advantage over linear probing, so the headline conclusion does not depend on the outliers. If anything, the case against accuracy strengthens—with the outliers removed, linear probing turns *positively* correlated with gFID (higher accuracy → worse generation), exactly the reversal reported in Sec. 2.

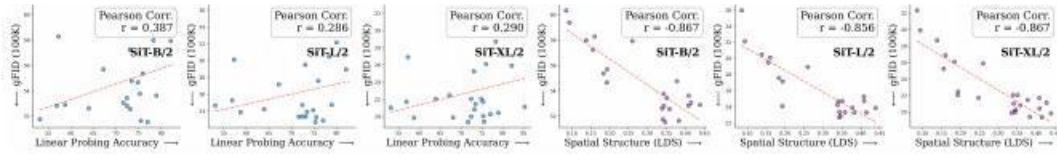


Figure 27: **Correlation analysis without outliers.** With the outliers (MoCOV3-L and MAE-L) removed, spatial structure still correlates strongly with generation performance at every model size (B, L, XL) (Pearson $|r| > 0.85$), while linear probing stays weak (Pearson $|r| < 0.38$)—the $> 2\times$ predictive gap is unchanged. Notably, without the outliers linear probing flips to a small *positive* correlation with gFID (higher accuracy → worse generation), reinforcing the accuracy-generation reversal from Sec. 2.

M.4 ADDITIONAL QUALITATIVE RESULTS

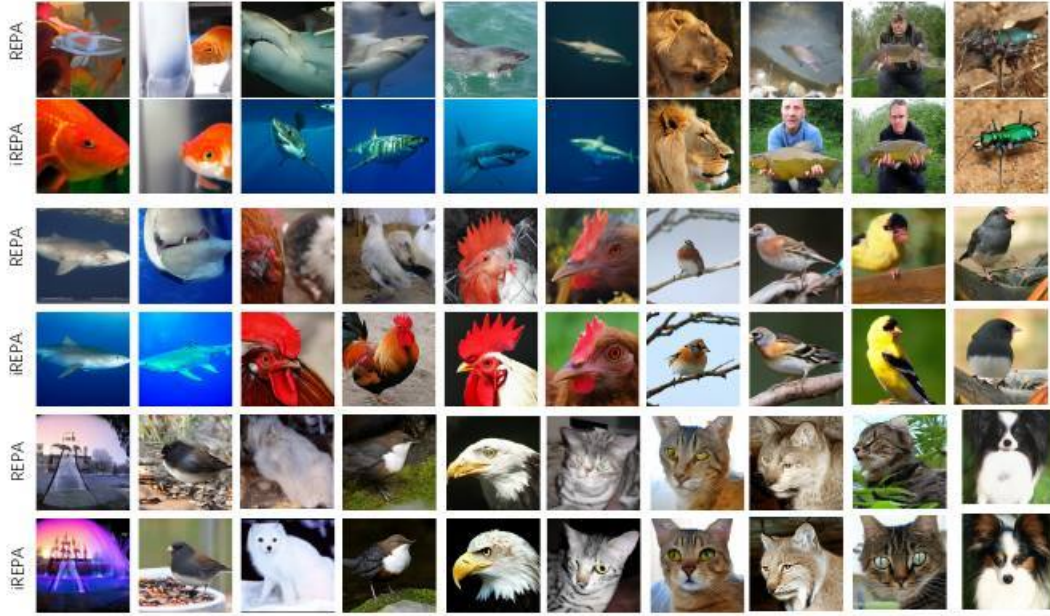


Figure 28: **Additional Qualitative Results** comparing generation outputs before (REPA) and after (iREPA) the spatial improvements, using PE-G (Bolya et al., 2025) as the pretrained vision encoder, 400K steps (80 epochs), and classifier-free guidance scale 4.0. The qualitative gap mirrors the quantitative one (§4): iREPA outputs are visibly sharper and more coherent.

M.5 ADDITIONAL RESULTS WITH ALTERNATIVE EVALUATION METRICS

A natural concern is whether our conclusions ride on the choice of FID. To rule this out, beyond the traditional metrics (Inception Score, FID, sFID, Precision, Recall) we re-evaluate both key findings under an alternative metric with distinct failure modes (Stein et al., 2023; Yang et al., 2023; Kynkäänniemi et al., 2023; Jayasumana et al., 2024). Specifically, we use the CMMD metric (Jayasumana et al., 2024) (PyTorch implementation of (Paul, 2024)) against the standard OpenAI ADM reference set (Dhariwal & Nichol, 2021), and re-check the results of both §3 and §4.

Spatial structure correlates far more strongly with generation performance than linear probing. Repeating the §3 correlation analysis across encoders with the CMMD metric (Jayasumana et al., 2024) (Figure 29; SiT-B/2, 100K steps, REPA) reproduces the same clean separation as under FID: every spatial metric is strongly predictive of CMMD (Pearson $|r| > 0.88$) while linear probing is essentially uninformative (Pearson $|r| = 0.074$). The predictive gap is even wider here—over $10\times$ —so the key empirical findings from §3 are robust to the choice of evaluation metric (FID or CMMD) and the ranking of the two predictors is not an artifact of FID.

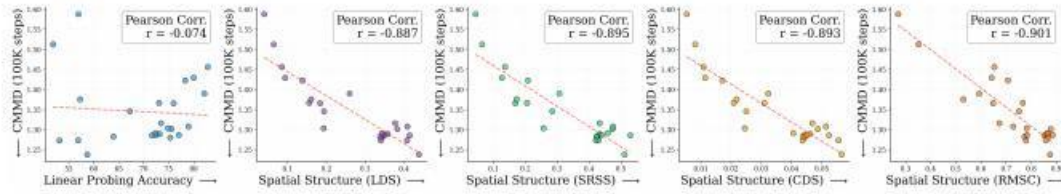


Figure 29: Correlation analysis with CMMD metric. The §3 head-to-head, now with CMMD (Jayasumana et al., 2024) in place of gFID (SiT-B/2, 100K steps, REPA). Every spatial metric is strongly predictive of CMMD (Pearson $|r| > 0.88$) while linear probing is not (Pearson $|r| = 0.074$)—a $> 10\times$ gap that mirrors the FID result and shows the finding is metric-independent.

Accentuating transfer of spatial features helps improve generation quality. We likewise re-check the §4 result under CMMD (Jayasumana et al., 2024) alongside the traditional metrics (gFID, sFID, IS, etc.). Table 8 confirms it: across encoders, +iREPA is the better row on CMMD and on the traditional metrics alike (IS, FID, sFID, Prec., Recall), and does so both with and without classifier-free guidance—so the improvement is not specific to any single metric.

Vision Encoder	Steps	w/o CFG						w/ CFG					
		CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑	CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑
DINOv2-B	100K	0.702	69.20	19.3	5.89	0.64	0.61	0.529	157.2	6.35	5.91	0.77	0.54
+iREPA	100K	0.652	77.92	16.9	6.26	0.66	0.61	0.484	179.3	5.15	6.23	0.78	0.54
DINOv2-B	400K	0.455	127.4	7.76	5.06	0.70	0.66	0.320	263.0	1.98	4.60	0.80	0.61
+iREPA	400K	0.438	128.6	7.52	4.89	0.71	0.65	0.310	268.8	1.93	4.59	0.80	0.60
DINOv3-B	100K	0.749	63.64	21.4	6.14	0.63	0.60	0.571	144.0	7.57	6.09	0.76	0.53
+iREPA	100K	0.651	78.79	16.2	6.14	0.66	0.61	0.481	181.9	4.87	6.10	0.78	0.55
DINOv3-B	400K	0.474	126.7	8.10	5.06	0.70	0.66	0.336	261.2	1.99	4.58	0.80	0.61
+iREPA	400K	0.441	132.9	7.13	4.93	0.71	0.66	0.314	272.4	1.89	4.58	0.80	0.60
WebSSL-1B	100K	0.825	53.87	25.5	6.57	0.61	0.59	0.622	124.0	9.59	6.37	0.76	0.52
+iREPA	100K	0.653	77.47	16.6	6.18	0.66	0.61	0.484	177.6	5.09	6.11	0.79	0.54
WebSSL-1B	400K	0.512	116.9	9.39	5.14	0.70	0.64	0.354	250.5	2.24	4.61	0.81	0.58
+iREPA	400K	0.445	130.8	7.48	4.91	0.70	0.65	0.311	271.6	1.90	4.58	0.80	0.61
PE-Core-G	100K	0.922	42.74	32.3	6.70	0.57	0.59	0.714	97.2	14.1	6.56	0.71	0.53
+iREPA	100K	0.697	75.01	18.1	6.03	0.64	0.61	0.525	176.8	5.66	6.08	0.77	0.54
PE-Core-G	400K	0.525	109.4	10.4	5.00	0.69	0.64	0.366	238.4	2.44	4.57	0.81	0.59
+iREPA	400K	0.458	132.0	7.78	5.02	0.70	0.65	0.322	275.4	1.93	4.59	0.80	0.61
CLIP-L	100K	0.790	54.07	25.2	6.65	0.61	0.60	0.605	124.5	9.77	6.59	0.75	0.52
+iREPA	100K	0.657	74.46	17.8	6.33	0.65	0.61	0.489	172.0	5.67	6.33	0.78	0.54
CLIP-L	400K	0.487	117.8	8.97	4.98	0.70	0.65	0.339	258.7	2.15	4.64	0.81	0.59
+iREPA	400K	0.442	130.1	7.42	5.03	0.71	0.65	0.307	271.8	1.96	4.64	0.80	0.61

Table 8: **Robustness of the iREPA gain to the choice of evaluation metric.** Each encoder block compares REPA (top) against +iREPA (highlighted), holding SiT-XL/2, batch size 256, and traditional REPA training fixed; CMMD (Jayasumana et al., 2024) is computed with the PyTorch implementation of (Paul, 2024) against the standard OpenAI ADM reference set (Dhariwal & Nichol, 2021). Reading down each pair, +iREPA is the better row on CMMD and on the conventional metrics (IS, FID, etc.) alike, doing so for every encoder and both with and without CFG. That the improvement survives a metric (CMMD) with entirely different failure modes from FID confirms the §4 findings are not an artifact of the evaluation metric.

M.6 SPATIAL IMPROVEMENTS WITH SAM2

Table 9 and Figure 30 probe spatial normalization (§4) on SAM2-S (46M) encoder features². Here the gain is present but small (Table 9), and the reason is mechanistic rather than a counterexample: spatial normalization works by removing the global component (mean of patch tokens) to sharpen spatial contrast, but SAM2 starts with almost no global component to remove (validation accuracy < 24%), so there is little contrast left to recover (Figure 30). The small margin here is thus itself consistent with the spatial-structure account.

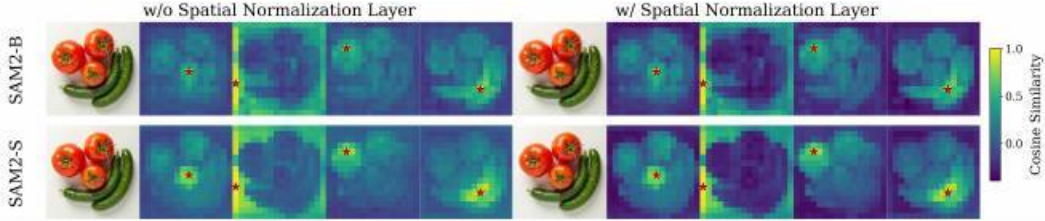


Figure 30: **Impact of spatial normalization on SAM2.** For SAM2, spatial normalization still raises the spatial contrast, but only slightly—and the small margin is expected, not anomalous. Spatial normalization (§4) works by removing the global component (mean of patch tokens); since SAM2 starts with almost no global component (validation accuracy < 24%), there is little to remove and hence little contrast to recover.

Vision Encoder	Steps	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
SAM2-S	100K	50.69	25.32	6.52	0.631	0.588
+iREPA	100K	53.14	24.52	6.28	0.629	0.591
SAM2-S	400K	110.77	9.54	4.93	0.699	0.638
+iREPA	400K	114.62	9.10	4.89	0.704	0.640

Table 9: **Impact of spatial improvements on SAM2.** Pairing REPA against +iREPA (highlighted) with SAM2-S (46M) as encoder, SiT-XL/2, without classifier-free guidance, at 100K and 400K steps. +iREPA is the better row on FID and IS at both checkpoints; the margins are modest because SAM2-S carries almost no global component to remove (< 24% accuracy), leaving spatial normalization little left to sharpen.

²Note that similar to (Bolya et al., 2025), we use the intermediate output of vision encoder for SAM2 features and not the mask logits. As shown in (Bolya et al., 2025), while mask logits lead to sharper spatial maps, the mask logits itself are not suitable as a target representation.

M.7 ADDITIONAL RESULTS WITH FULL-FINETUNING ACCURACY INSTEAD OF LINEAR PROBING ACCURACY

Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. To rule out that our conclusion hinges on the specific global-information probe, we swap linear probing for full-finetuning validation accuracy³ and repeat the §3 correlation analysis (Fig. 31; SiT-XL/2, 100K steps, REPA; full-finetuning at learning rate $5e-5$, warmup ratio 0.1, 30 epochs). The outcome is the same: every spatial metric correlates with generation far more strongly than the finetuning accuracy does. In fact gFID shows a weak *positive* correlation with finetuning accuracy (Pearson $r = 0.317$)—higher accuracy accompanying *worse* generation—the same accuracy-generation reversal documented in §2.

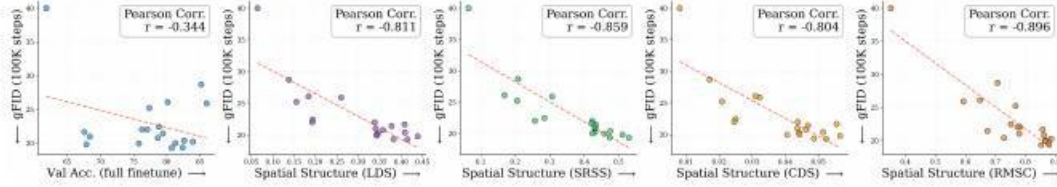


Figure 31: Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. The §3 head-to-head, using full-finetuning validation accuracy in place of linear probing (SiT-XL/2, 100K steps, REPA). Every spatial metric again correlates with generation far more strongly than the accuracy measure. gFID even shows a weak *positive* correlation with finetuning accuracy (Pearson $r = 0.317$)—higher accuracy, worse generation—matching the accuracy-generation reversal in §2.

³Please note that while the final findings remain similar, in context of REPA linear probing might be more accurate for estimating global information in “pretrained” vision encoders. This is because REPA uses the “pretrained” encoder features themselves for regularization. Finetuning the encoder itself can impact the amount of global information. Same as REPA (Yu et al., 2024), we therefore use linear probing accuracy as default for measuring global information.

M.8 ADDITIONAL RESULTS ON PIXEL-SPACE DIFFUSION MODELS

Method	#Params	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
PixelFlow (Chen et al., 2025)	459M	24.67	54.33	9.71	-	0.58
PixDDT (Wang et al., 2025b)	434M	36.24	46.37	17.14	-	0.63
PixNerd (Wang et al., 2025a)	458M	43.01	37.49	10.65	-	0.62
DeCo w/o $\mathcal{L}_{\text{FreqFM}}$	426M	46.44	34.12	10.41	-	0.64
DeCo + REPA (Ma et al., 2025b)	426M	48.35	31.35	9.34	-	0.65
JiT (Li & He, 2025a)	459M	29.37	49.06	11.21	0.40	0.62
JiT+iREPA (Ours)	459M	50.72	29.19	9.42	0.51	0.65

Table 10: **Generation performance of pixel-space diffusion models.** At matched 200K iterations, batch size 256, and 50-step Euler sampling (no CFG), adding iREPA to JiT (Li & He, 2025a) improves FID from 49.06 to 29.19 and IS from 29.37 to 50.72—and this simple recipe edges out the strongest reported pixel-space baseline, DeCo + REPA (Ma et al., 2025b) (FID 31.35), while also leading on precision and recall. DeCo numbers are taken directly from (Ma et al., 2025b).